Module #2:
The Theory of Sets

Rosen 8th ed., § § 2.1-2.2

2 Basic Structures: Sets, Functions, Sequences , Sums and Matrices

- **2.1** Sets
  2.2 Set Operations
  2.3 Functions
  2.4 Sequences and Summations
- 2.5 Cardinality of Sets
- 2.6 Matrices

Introduction to Set Theory
集合论(§2.1)

- A *set* is a new type of structure, representing an *unordered* collection (group, plurality) of zero or more *distinct* (different) objects.
- Set theory deals with operations between, relations among, and statements about sets.
- Sets are ubiquitous in computer software systems.
- *All* of mathematics can be defined in terms of some form of set theory (using predicate logic).
- https://zhuanlan.zhihu.com/p/38392100

Naïve set theory

- **Basic premise:** Any collection or class of objects (*elements*) that we can *describe* (by any means whatsoever) constitutes a set.
- But, the resulting theory turns out to be *logically inconsistent*!
  - This means, there exist naïve set theory propositions p such that you can prove that both p and $\neg p$ follow logically from the axioms of the theory!
  - $\therefore$ The conjunction of the axioms is a contradiction!
  - This theory is fundamentally uninteresting, because any possible statement in it can be (very trivially) "proved" by contradiction!
- More sophisticated set theories fix this problem.

Basic notations for sets

- For sets, we'll use variables *S*, *T*, *U*, …
- We can denote a set *S* in writing by listing all of its elements in curly braces:
  - {a, b, c} is the set of whatever 3 objects are denoted by a, b, c.
- *Set builder notation*: For any proposition *P*(*x*) over any universe of discourse, {*x*|*P*(*x*)} is *the set of all x such that P(x).*

Example

- The set V of all vowels in English alphabet can be written as V={*a,e,i,o,u*}.
- Speech sounds are generally divided into vowels and consonants.
- The set O of odd positive integers less than 10 can be expressed by O={1,3,5,7,9}.
- The set of positive integers less than 100 can be denoted by {1,2,3,…,99}.

Basic properties of sets

- Sets are inherently *unordered*:
  - No matter what objects a, b, and c denote, {a, b, c} = {a, c, b} = {b, a, c} = {b, c, a} = {c, a, b} = {c, b, a}.
- All elements are *distinct* (unequal); multiple listings make no difference!
  - If a=b, then {a, b, c} = {a, c} = {b, c} = {a, a, b, a, b, c, c, c, c}.
  - This set contains (at most) 2 elements!

Definition of Set Equality
集合相等

- Two sets are declared to be equal *if and only if* they contain exactly the same elements.
- In particular, it does not matter *how the set is defined or denoted.*
- **For example:** The set {1, 2, 3, 4} =
  {x | x is an integer where x>0 and x<5 } =
  {x | x is a positive integer whose square is >0 and <25}

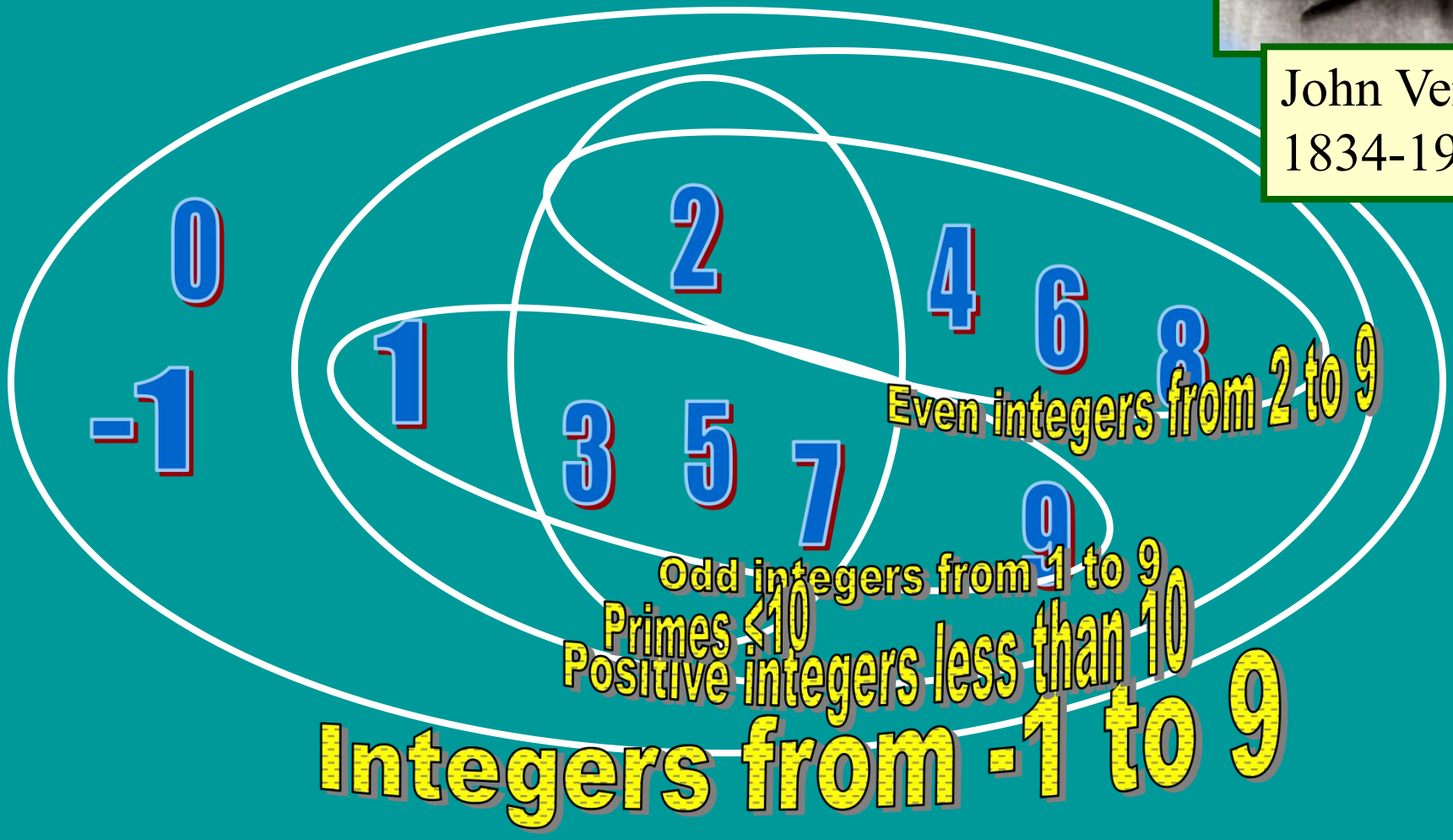
Infinite Sets无限集

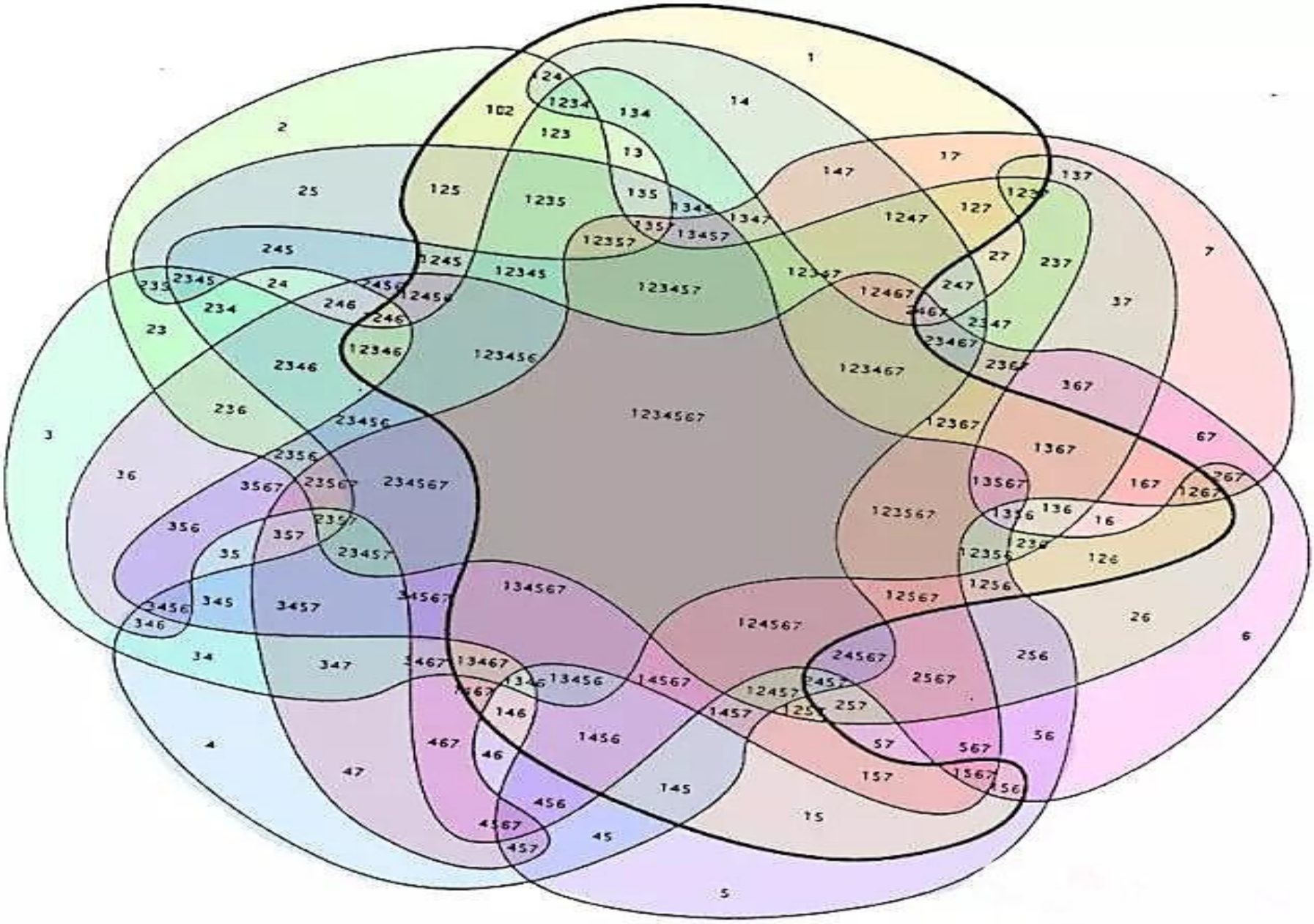
- Conceptually, sets may be *infinite* (*i.e.,* not *finite*, without end, unending).
- Symbols for some special infinite sets:
  N = {0, 1, 2, …} The **N**atural numbers.
  Z = {…, -2, -1, 0, 1, 2, …} The **Z**ntegers.
  R = The "**R**eal" numbers, such as 374.18284719294981819172819431 25…
- "Blackboard Bold" or double-struck font (ℕ ,ℤ ,ℝ) is also often used for these special number sets.
- Infinite sets come in different sizes!

Venn Diagrams
文氏图



John Venn
1834-1923




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Basic Set Relations:Member of 成员

- $x \in S$ ("x is in S") is the proposition that object x is an $\in$*lement* or *member* of set S.
  - *e.g.* $3 \in \mathbf{N}$, "a" $\in \{x \mid x$ is a letter of the alphabet$\}$
  - Can define set equality in terms of $\in$ relation:
    $\forall S,T: S=T \leftrightarrow (\forall x: x \in S \leftrightarrow x \in T)$
    "Two sets are equal iff they have all the same members."
- $x \notin S :\equiv \neg(x \in S)$ "x is not in S"

The Empty Set空集

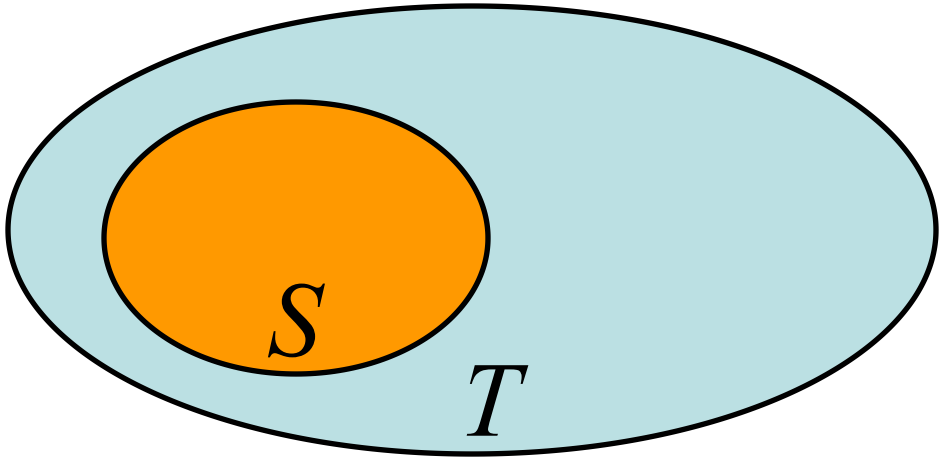
- $\varnothing$ (“null”, “the empty set”) is the unique set that contains no elements whatsoever.
- $\varnothing = \{\} = \{x|\mathbf{False}\}$
- No matter the domain of discourse, we have the axiom $\neg\exists x: x \in \varnothing$.

Subset子集

- $S \subseteq T$ ("S is a subset of T") means that every element of S is also an element of T.
- $S \subseteq T \Leftrightarrow \forall x\,(x \in S \rightarrow x \in T)$
- $\varnothing \subseteq S$, $S \subseteq S$.

Proper (Strict) Subsets真子集

- $S \subset T$ ("S is a proper subset of T") means that $S \subseteq T$ but $T \not\subseteq S$.



Venn Diagram equivalent of $S \subset T$

Example:
$\{1,2\} \subset \{1,2,3\}$

Sets Are Objects, Too!

- The objects that are elements of a set may *themselves* be sets.
- *E.g.* let $S=\{x \mid x \subseteq \{1,2,3\}\}$
  then $S=\{\varnothing,$
  $\{1\}, \{2\}, \{3\},$
  $\{1,2\}, \{1,3\}, \{2,3\},$
  $\{1,2,3\}\}$
- Note that $1 \neq \{1\} \neq \{\{1\}\}$!!!!



Cardinality and Finiteness基数

- $|S|$ (read "the *cardinality* of S") is a measure of how many different elements S has.
- *E.g.*, $|\varnothing|=0$, $|\{1,2,3\}| = 3$, $|\{a,b\}| = 2$, $|\{\{1,2,3\},\{4,5\}\}| =$ 2
- If $|S| \in \mathbf{N}$, then we say S is *finite*. Otherwise, we say S is *infinite*.
- What are some infinite sets we've seen?

$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{R}$

The *Power Set* 冪集Operation

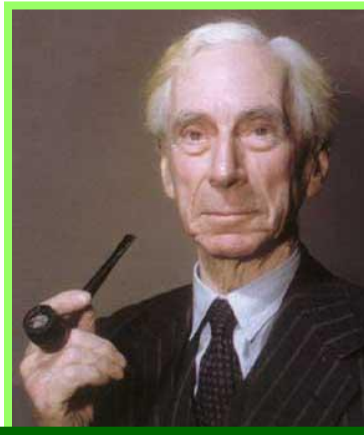
- The *power set* P(*S*) of a set *S* is the set of all subsets of *S*. $P(S) :\equiv \{x \mid x \subseteq S\}$.
- *E.g.* P({a,b}) = {∅, {a}, {b}, {a,b}}.
- Sometimes P(*S*) is written $\mathbf{2}^S$. Note that for finite *S*, $|P(S)| = 2^{|S|}$.
- It turns out $\forall S: |P(S)| > |S|$, *e.g.* $|P(\mathbf{N})| > |\mathbf{N}|$. *There are different sizes of infinite sets*!

Review: Set Notations So Far

- Variable objects x, y, z; sets S, T, U.
- Literal set {a, b, c} and set-builder $\{x|P(x)\}$.
- $\in$ relational operator, and the empty set $\varnothing$.
- Set relations $=$, $\subseteq$, $\supseteq$, $\subset$, $\supset$, $\not\subset$, etc.
- Venn diagrams.
- Cardinality $|S|$ and infinite sets **N**, **Z**, **R**.
- Power sets $\mathrm{P}(S)$.

Naïve Set Theory is Inconsistent

- There are some naïve set *descriptions* that lead to pathological structures that are not *well-defined*.
  - (That do not have self-consistent properties.)
- These "sets" mathematically *cannot* exist.
- *E.g.* let $S = \{x \mid x \notin x\}$. Is $S \in S$?
- Therefore, consistent set theories must restrict the language that can be used to describe sets.
- For purposes of this class, don't worry about it!



Bertrand Russell
1872-1970

- 1901年，B.Russell发现了朴素集合论的悖论：**罗素悖论**.

> 设 M 为一集合， $P(M)$ 表示“ M 是不以自己作为元素的集合”这种性质.
> 下面考察具有性质 P 的类 $K = \{M|P(M)\}$.
> 如果 K 是集合，由排中律， $P(K)$ 或 $\neg P(K)$ 仅有一个为真.
> 考察 $P(K)$ ，由 K 的定义， K 不以自己为元素，因此 $P(K)$ 为假.
> 考察 $\neg P(K)$ ，若其成立，意味着 K 以自己为元素，这又与 K 的定义（ K 不以自己为元素）相矛盾.
> 因此K不是集合.而此时，“任何性质都定义一个具有该性质的事物的集合.”不被满足.

- 1908年，提出了两种避免悖论的方法，罗素的Type theory和Zermelo集合论.
- 而现在最广泛使用和接受的集合论是ZFC，它由Zermelo-Fraenkel集合论和附加的选择公理组成.

Ordered *n*-tuples 有序n元组

- These are like sets, except that duplicates matter, and the order makes a difference. Ordered pairs 序偶
- For $n \in \mathbf{N}$, an *ordered n-tuple* or a *sequence* or *list of length n* is written $(a_1, a_2, \ldots, a_n)$. Its *first* element is a_1, *etc*
- Note that $(1, 2) \neq (2, 1) \neq (2, 1, 1)$.
- Empty sequence, singlets, pairs, triples, quadruples, quintuples, …, *n*-tuples.

Contrast with sets' {}

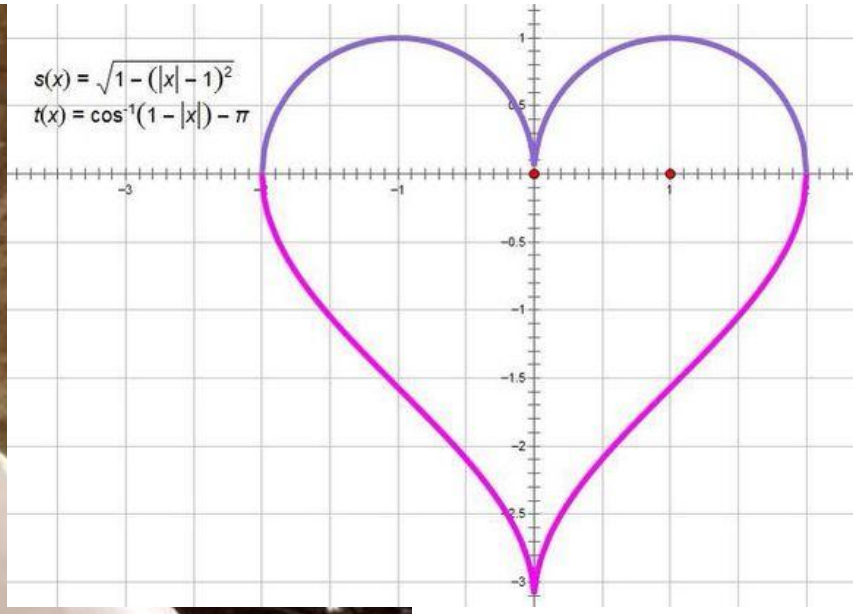
Cartesian Products of Sets
迪卡尔集

- For sets A, B, their *Cartesian product* $A\times B :\equiv \{(a, b) \mid a\in A \wedge b\in B\}$.
- *E.g.* {a,b}×{1,2} = {(a,1),(a,2),(b,1),(b,2)}
- Note that for finite A, B, $|A\times B|=|A||B|$.
- Note that the Cartesian product is *not* commutative: *i.e.*, $\neg\forall AB: A\times B=B\times A$.
- Extends to $A_1 \times A_2 \times \ldots \times A_n$...



René Descartes
(1596-1650)







- 1650年，斯德哥尔摩的街头，52岁的笛卡尔邂逅了18岁的瑞典公主克里斯汀。　　那时，落魄、一文不名的笛卡尔过着乞讨的生活，全部的财产只有身上穿的破破烂烂的衣服和随身所带的几本数学书籍。生性清高的笛卡尔从来不开口请求路

Definition of relations

- Let *A* and *B* be two sets. A *binary relation R from A to B* is a subset of $A \times B$.
  - Note that the order of the two sets matters.
- More generally, let A_1, A_2, ..., A_n be *n* sets. An *n-ary relation R* on these sets is a subset of $A_1 \times A_2 \times ... \times A_n$.
  - The sets A_i are known as the *domains* of the relation, and *n* as its *degree*
  - Again, the order of the domains matters.

Review of § 2.1

- Sets S, T, U… Special sets **N**, **Z**, **R**.
- Set notations {a,b,...}, {x|P(x)}…
- Set relation operators $x \in S$, $S \subseteq T$, $S \supseteq T$, $S = T$, $S \subset T$, $S \supset T$. (These form propositions.)
- Finite vs. infinite sets.
- Set operations $|S|$, P(S), $S \times T$.

Start §2.2: 集合运算The Union Operator并

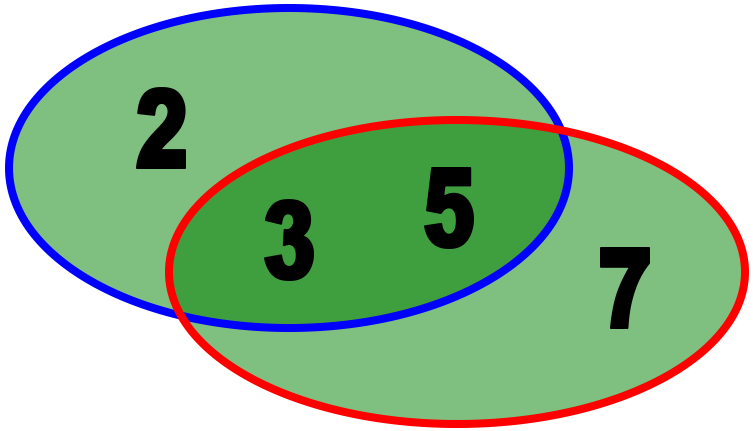
- For sets A, B, their ∪*nion* $A \cup B$ is the set containing all elements that are either in A, **or** ("$\vee$") in B (or, of course, in both).
- Formally, $\forall A,B: A \cup B = \{x \mid x \in A \vee x \in B\}$.
- Note that $A \cup B$ is a **superset** of both A and B (in fact, it is the smallest such superset):

  $\forall A, B: (A \cup B \supseteq A) \wedge (A \cup B \supseteq B)$

Union Examples

- {a,b,c}∪{2,3} = {a,b,c,2,3}
- {2,3,5}∪{3,5,7} = {2,3,5,3,5,7} = {2,3,5,7}

Required Form



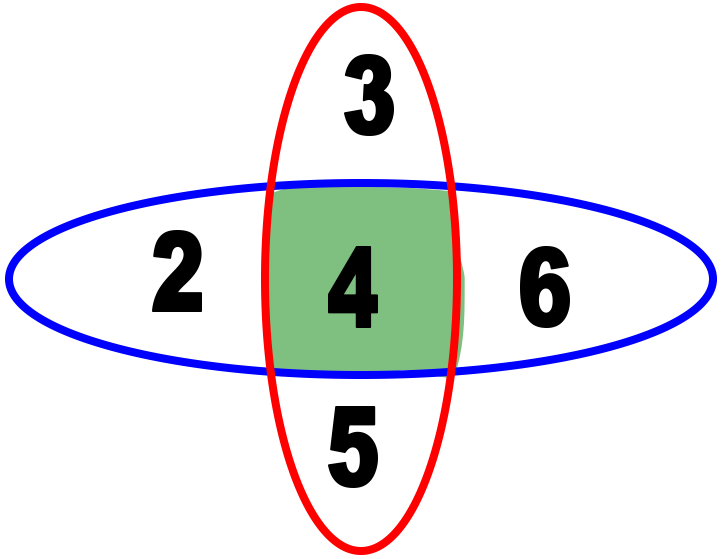
The Intersection Operator交

- For sets *A*, *B*, their *intersection* $A \cap B$ is the set containing all elements that are simultaneously in *A* **and** ("$\wedge$") in *B*.
- Formally, $\forall A,B: A \cap B = \{x \mid x \in A \wedge x \in B\}$.
- Note that $A \cap B$ is a **subset** of both A and B (in fact it is the largest such subset):

$$\forall A, B: (A \cap B \subseteq A) \wedge (A \cap B \subseteq B)$$

Intersection Examples

- {a,b,c}∩{2,3} = ∅
- {2,4,6}∩{3,4,5} = {4}



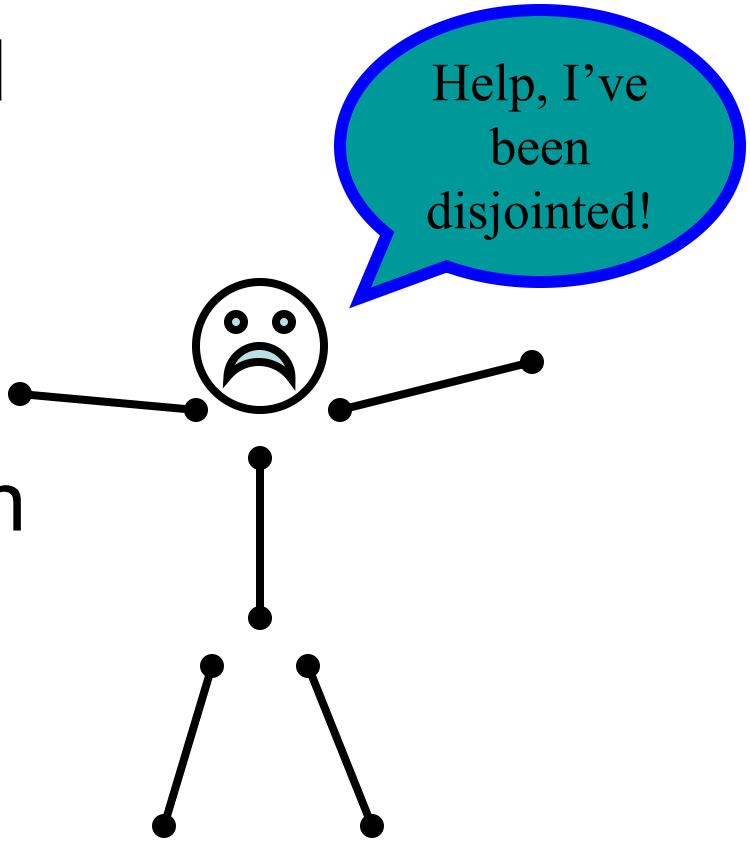




A Venn diagram showing a union of the Chinese tea set and the Green tea set. At the intersection of these sets are different types of Chinese green tea.

Disjointedness 互斥

- Two sets *A*, *B* are called *disjoint* (*i.e.*, unjoined) iff their intersection is empty. ($A \cap B = \varnothing$)
- Example: the set of even integers is disjoint with the set of odd integers.



Inclusion-Exclusion Principle
容斥原理

- How many elements are in $A \cup B$?

Subtract out items in intersection, to compensate for double-counting them!

- Example: How many students are on
  $\cup M$,
  I = {s | s sent the TA information sheet}
  M = {s | s sent the TA
  Some students did both!

$$|E| = |I \cup M| = |I| + |M| - |I \cap M|$$

Set Difference差

- For sets *A*, *B*, the *difference of A and B*, written *A*–*B*, is the set of all elements that are in *A* but not *B*. Formally:

$$A - B :\equiv \{x \mid x \in A \wedge x \notin B\}$$
$$= \{x \mid \neg(x \in A \rightarrow x \in B)\}$$

- Also called:
  The *complement of B with respect to A*.

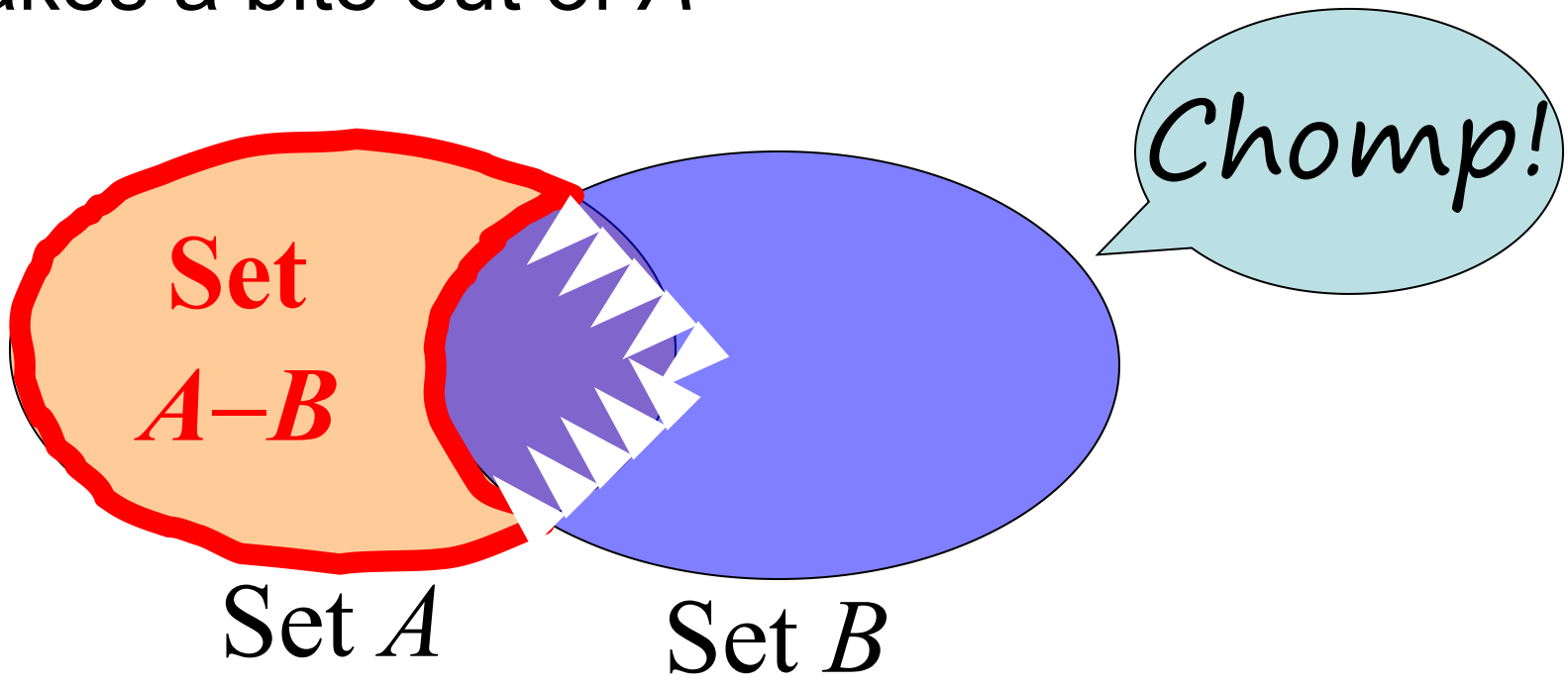
Set Difference Examples



- {1,2,3,4,5,6} – {2,3,5,7,9,11} =
  {1,4,6}
- **Z** – **N** = {… , −1, 0, 1, 2, … } – {0, 1, … }
  = {*x* | *x* is an integer but not a nat. #}
  = {*x* | *x* is a negative integer}
  = {… , −3, −2, −1}

Set Difference - Venn Diagram

- *A*−*B* is what's left after *B* "takes a bite out of *A*"



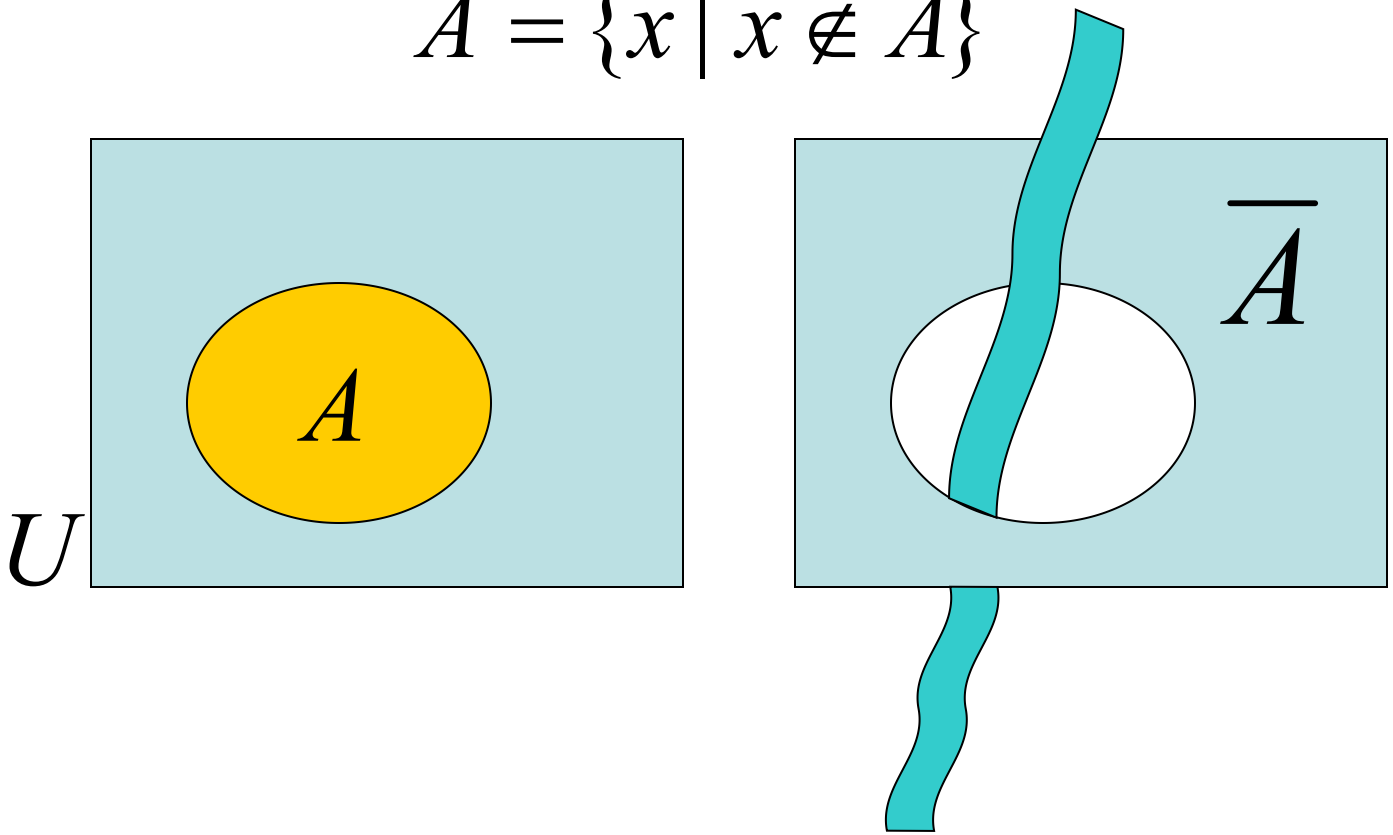
Set Complements补

- The *universe of discourse* can itself be considered a set, call it *U*.
- When the context clearly defines *U*, we say that for any set $A \subseteq U$, the *complement* of *A*, written $\overline{A}$, is the complement of *A* w.r.t. *U*, *i.e.*, it is *U*–*A*.
- *E.g.,* If *U*=**N**, $\overline{\{3,5\}} = \{0,1,2,4,6,7,...\}$

More on Set Complements

- An equivalent definition, when *U* is clear:

$$\overline{A} = \{x \mid x \notin A\}$$



Set Identities特性

- Identity: $A \cup \varnothing = A = A \cap U$
- Domination: $A \cup U = U$, $A \cap \varnothing = \varnothing$
- Idempotent: $A \cup A = A = A \cap A$
- Double complement: $\overline{(\overline{A})} = A$
- Commutative: $A \cup B = B \cup A$, $A \cap B = B \cap A$
- Associative: $A \cup (B \cup C) = (A \cup B) \cup C$,
  $A \cap (B \cap C) = (A \cap B) \cap C$

DeMorgan's Law for Sets

- Exactly analogous to (and provable from) DeMorgan's Law for propositions.

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

Proving Set Identities

To prove statements about sets, of the form $E_1 = E_2$ (where the Es are set expressions), here are three useful techniques:

1. Prove $E_1 \subseteq E_2$ and $E_2 \subseteq E_1$ separately.
2. Use set builder notation & logical equivalences.
3. Use a *membership table*.

Method 1: Mutual subsets

Example: Show $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

- Part 1: Show $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.
  - Assume $x \in A \cap (B \cup C)$, & show $x \in (A \cap B) \cup (A \cap C)$.
  - We know that $x \in A$, and either $x \in B$ or $x \in C$.
    - Case 1: $x \in B$. Then $x \in A \cap B$, so $x \in (A \cap B) \cup (A \cap C)$.
    - Case 2: $x \in C$. Then $x \in A \cap C$, so $x \in (A \cap B) \cup (A \cap C)$.
  - Therefore, $x \in (A \cap B) \cup (A \cap C)$.
  - Therefore, $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.
- Part 2: Show $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$. …

Method 3: Membership Tables

- Just like truth tables for propositional logic.
- Columns for different set expressions.
- Rows for all combinations of memberships in constituent sets.
- Use “1” to indicate membership in the derived set, “0” for non-membership.
- Prove equivalence with identical columns.

Membership Table Example

Prove $(A \cup B) - B = A - B$.

A	B	$A \cup B$	$(A \cup B) - B$	$A - B$
0	0	0	0	0
0	1	1	0	0
1	0	1	1	1
1	1	1	0	0

Membership Table Exercise

Prove $(A \cup B) - C = (A - C) \cup (B - C)$.

A	B	C	$A \cup B$	$(A \cup B) - C$	$A - C$	$B - C$	$(A - C) \cup (B - C)$
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

Review of § 2.1-2.2

- Sets S, T, U… Special sets **N**, **Z**, **R**.
- Set notations {a,b,...}, {x|P(x)}…
- Relations $x \in S$, $S \subseteq T$, $S \supseteq T$, $S = T$, $S \subset T$, $S \supset T$.
- Operations |S|, P(S), $\times$, $\cup$, $\cap$, $\overline{S}$
- Set equality proof techniques:
  - Mutual subsets.
  - Derivation using logical equivalences.

Generalized Unions & Intersections

- Since union & intersection are commutative and associative, we can extend them from operating on *ordered pairs* of sets (A,B) to operating on sequences of sets $(A_1,\ldots,A_n)$, or even on unordered *sets* of sets, $X=\{A \mid P(A)\}$.

Generalized Union

- Binary union operator: $A \cup B$
- *n*-ary union:
  $A \cup A_2 \cup \ldots \cup A_n :\equiv ((\ldots((A_1 \cup A_2) \cup \ldots) \cup A_n)$
  (grouping & order is irrelevant)
- "Big U" notation: $\bigcup_{i=1}^{n} A_i$
- Or for infinite sets of sets: $\bigcup_{A \in X} A$

Generalized Intersection

- Binary intersection operator: $A \cap B$
- *n*-ary intersection:
  $A_1 \cap A_2 \cap \ldots \cap A_n \equiv ((\ldots((A_1 \cap A_2) \cap \ldots) \cap A_n)$
  (grouping & order is irrelevant)
- "Big Arch" notation: $\bigcap_{i=1}^{n} A_i$
- Or for infinite sets of sets: $\bigcap_{A \in X} A$

Representations

- A frequent theme of this course will be methods of *representing* one discrete structure using another discrete structure of a different type.
- *E.g.*, one can represent natural numbers as
  - Sets: **0**:≡∅, **1**:≡{**0**}, **2**:≡{**0**,**1**}, **3**:≡{**0**,**1**,**2**}, …
  - Bit strings:
    0:≡`0`, **1**:≡`1`, **2**:≡`10`, **3**:≡`11`, **4**:≡`100`, …

Representing Sets with Bit Strings

For an enumerable u.d. U with ordering $x_1, x_2, \ldots$, represent a finite set $S \subseteq U$ as the finite bit string $B=b_1 b_2 \ldots b_n$ where $\forall i: x_i \in S \leftrightarrow (i<n \wedge b_i=1)$.

E.g. $U=\mathbf{N}$, $S=\{2,3,5,7,11\}$, $B=001101010001$.

In this representation, the set operators "$\cup$", "$\cap$", "$\overline{\ }$" are implemented directly by bitwise OR, AND, NOT!

EXAMPLE 20： The bit strings for the sets {1, 2, 3, 4, 5} and {1, 3, 5, 7, 9} are 11 1110 0000 and 10 1010 1010, respectively. Use bit strings to find the union and intersection of these sets.

- Solution: The bit string for the union of these sets is

11 1110 0000 $\vee$ 10 1010 1010 = 11 1110 1010,

which corresponds to the set {1, 2, 3, 4, 5, 7, 9}.

- The bit string for the intersection of these sets is

11 1110 0000 $\wedge$ 10 1010 1010 = 10 1010 0000,

which corresponds to the set {1, 3, 5}.

Multisets

- A **multiset** (short for multiple-membership set) is an unordered collection of elements where an element can occur as a member more than once.

$\{4 \cdot a, 1 \cdot b, 3 \cdot c\}$
and $\{3 \cdot a, 4 \cdot b, 2 \cdot d\}$,

- The notation $\{m_1 \cdot a_1, m_2 \cdot a_2, \ldots, m_r \cdot a_r\}$ denotes the multiset with element $a1$ occurring m_1 times, element a_2 occurring m_2 times, and so on.
- The numbers m_i, $i = 1, 2, \ldots, r$, are called the **multiplicities** of the elements a_i, $i = 1, 2, \ldots, r$.(Elements not in a multiset are assigned 0 as their multiplicity in this set.)
- The cardinality of a multiset is defined to be the sum of the multiplicities of its elements.

Let *P* and *Q* be multisets. The **union** of the multisets *P* and *Q* is the multiset in which the multiplicity of an element is the maximum of its multiplicities in *P* and *Q*.

The **intersection** of *P* and *Q* is the multiset in which the multiplicity of an element is the minimum of its multiplicities in *P* and *Q*.

The **difference** of *P* and *Q* is the multiset in which the multiplicity of an element is the multiplicity of the element in *P* less its multiplicity in *Q* unless this difference is negative, in which case the multiplicity is 0.

- The **sum** of P and Q is the multiset in which the multiplicity of an element is the sum of multiplicities in P and Q.
- The union, intersection, and difference of P and Q are denoted by $P \cup Q$, $P \cap Q$, and $P - Q$, respectively (where these operations should not be confused with the analogous operations for sets). The sum of P and Q is denoted by $P + Q$.

作业

- 2.1　8,12,18,24,34,48
- 2.2　4,21,32,54,58,62

2 Basic Structures: Sets, Functions, Sequences , Sums and Matrices

- **2.1** Sets
  2.2 Set Operations
  2.3 Functions
  2.4 Sequences and Summations
- 2.5 Cardinality of Sets
- 2.6 Matrices

Module #3:
Functions

Rosen 8th ed., § 2.3

On to section 2.3… Functions

- From calculus, you are familiar with the concept of a real-valued function f, which assigns to each number $x \in \mathbf{R}$ a particular value $y=f(x)$, where $y \in \mathbf{R}$.
- But, the notion of a function can also be naturally generalized to the concept of assigning elements of *any* set to elements of *any* set. (Also known as a *map*.)

Function: Formal Definition

- For any sets A, B, we say that a *function f from (or "mapping") A to B* ($f{:}A \to B$) is a particular assignment of exactly one element $f(x) \in B$ to each element $x \in A$.
- Some further generalizations of this idea:
  - A *partial* (non-*total*) function *f* assigns *zero or one* elements of B to each element $x \in A$.
  - Functions of *n* arguments; relations (ch. 9).

Functions

- A function f: $A \rightarrow B$ can also be defined as a subset of $A \times B$ (a relation). This subset is restricted to be a relation where no two elements of the relation have the same first element.
- Specifically, a function f from A to B contains one, and only one ordered pair (a, b) for every element $a \in A$.

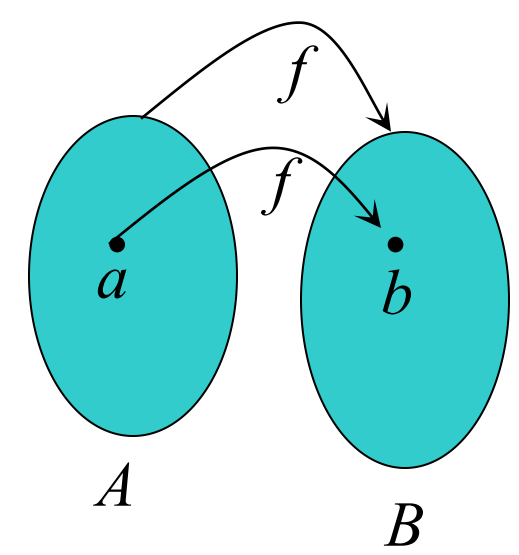
$$\forall x[x \in A \rightarrow \exists y[y \in B \wedge (x, y) \in f]]$$

and

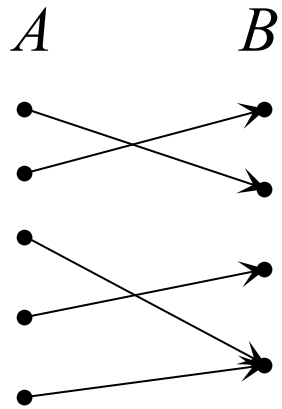
$$\forall x, y_1, y_2[[(x, y_1) \in f \wedge (x, y_2)] \rightarrow y_1 = y_2]$$

Graphical Representations

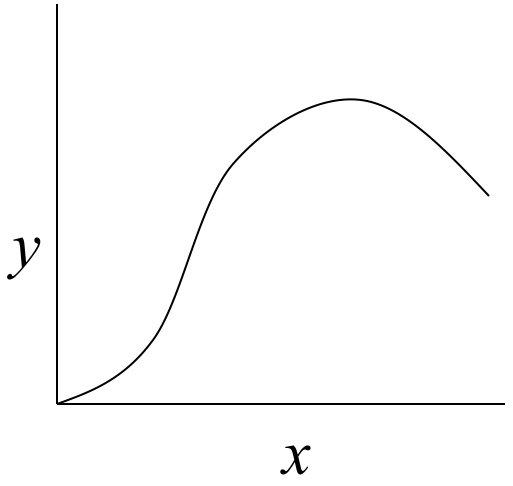
- Functions can be represented graphically in several ways:



Like Venn diagrams



Bipartite Graph



Plot

Functions We've Seen So Far

- A *proposition* can be viewed as a function from "situations" to truth values {**T**,**F**}
  - A logic system called *situation theory*.
  - *p*="It is raining."; *s*=our situation here,now
  - $p(s) \in \{\mathbf{T},\mathbf{F}\}$.
- A *propositional operator* can be viewed as a function from *ordered pairs* of truth values to truth values: *e.g.*, $\vee((\mathbf{F},\mathbf{T})) = \mathbf{T}$.

Another example: $\rightarrow((\mathbf{T},\mathbf{F})) = \mathbf{F}$.

More functions so far…

- A *predicate* can be viewed as a function from *objects* to *propositions* (or truth values): *P* :≡ "is 7 feet tall";
  P(Mike) = "Mike is 7 feet tall." = **False**.
- A *bit string B of length n* can be viewed as a function from the numbers {1,…,*n*} (bit *positions*) to the *bits* {0,1}.
  E.g., *B*=101 ➔ *B*(3)=1.

Still More Functions

- A *set S* over universe *U* can be viewed as a function from the elements of *U* to {**T**, **F**}, saying for each element of *U* whether it is in *S*. *S*={3}➔ *S*(0)=**F**, *S*(3)=**T**.
- A *set operator* such as $\cap$,$\cup$, $\bar{}$ can be viewed as a function from pairs of sets to sets.
  - Example: $\cap(\{\{1,3\},\{3,4\}\}) = \{3\}$

A Neat Trick

- Sometimes we write Y^X to denote the set F of *all* possible functions $f{:}X\to Y$.
- This notation is especially appropriate, because for finite X, Y, we have $|F| = |Y|^{|X|}$.
- If we use representations $\mathbf{F}\equiv\mathbf{0}$, $\mathbf{T}\equiv\mathbf{1}$, $\mathbf{2}:\equiv\{\mathbf{0},\mathbf{1}\}=\{\mathbf{F},\mathbf{T}\}$, then a subset $T\subseteq S$ is just a function from S to $\mathbf{2}$, so the power set of S (set of all such fns.) is $\mathbf{2}^S$ in this notation.

Some Function Terminology

- If it is written that $f{:}A \to B$, and $f(a)=b$ (where $a \in A$ & $b \in B$), then we say:
  - A is the *domain* of f.
  - B is the *codomain* of f.
  - b is the *image* of a under f.
  - a is a *pre-image* of b under f.
    - In general, b may have more than 1 pre-image.
  - The *range* $R \subseteq B$ of f is $R=\{b \mid \exists a\ f(a)=b\}$.

We also say the *signature* of f is $A \to B$.

Range versus Codomain

- The range of a function might *not* be its whole codomain.
- The codomain is the set that the function is *declared* to map all domain values into.
- The range is the *particular* set of values in the codomain that the function *actually* maps elements of the domain to.

Images of Sets under Functions

- Given $f{:}A \to B$, and $S \subseteq A$,
- The *image* of S under f is simply the set of all images (under f) of the elements of S.

$$f(S) :\equiv \{f(s) \mid s \in S\}$$
$$:\equiv \{b \mid \exists\, s \in S{:}\ f(s)=b\}.$$

- Note the range of f can be defined as simply the image (under f) of f's domain!

Range vs. Codomain - Example

- Suppose I declare to you that: "*f* is a function mapping students in this class to the set of grades {A,B,C,D,E}."
- At this point, you know *f*'s codomain is: {A,B,C,D,E}, and its range is unknown!.
- Suppose the grades turn out all As and Bs.
- Then the range of *f* is {A,B}, but its codomain is still {A,B,C,D,E}!.

Operators (general definition)

- An *n*-ary *operator over* (or *on*) the set *S* is any function from the set of ordered *n*-tuples of elements of *S*, to *S* itself.
- *E.g.*, if *S*={**T**,**F**}, $\neg$ can be seen as a unary operator, and $\wedge$,$\vee$ are binary operators on *S*.
- Another example: $\cup$ and $\cap$ are binary operators on the set of all sets.

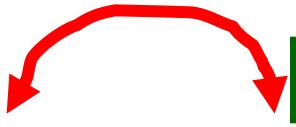
Constructing Function Operators

- If $\bullet$ ("dot") is any operator over B, then we can extend $\bullet$ to also denote an operator over functions $f{:}A \to B$.
- *E.g.*: Given any binary operator $\bullet{:}B \times B \to B$, and functions $f,g{:}A \to B$, we define $(f \bullet g){:}A \to B$ to be the function defined by: $\forall a \in A$, $(f \bullet g)(a) = f(a) \bullet g(a)$.

Function Operator Example

- $+, \times$ ("plus","times") are binary operators over **R**. (Normal addition & multiplication.)
- Therefore, we can also add and multiply *functions* f,g:$\mathbf{R}\to\mathbf{R}$:
  - $(f+g)$:$\mathbf{R}\to\mathbf{R}$, where $(f+g)(x) = f(x)+g(x)$
  - $(f \times g)$:$\mathbf{R}\to\mathbf{R}$, where $(f \times g)(x) = f(x) \times g(x)$

Function Composition Operator



- For functions $g{:}A \to B$ and $f{:}B \to C$, there is a special operator called *compose* (" $\circ$ ").
  - It composes (creates) a new function out of f and g by applying f to the result of applying g.
  - We say $(f \circ g){:}A \to C$, where $(f \circ g)(a) :\equiv f(g(a))$.
  - Note $g(a) \in B$, so $f(g(a))$ is defined and $\in C$.
  - Note that $\circ$ (like Cartesian $\times$, but unlike $+, \wedge, \cup$) is non-commuting. (Generally, $f \circ g \neq g \circ f$.)

One-to-One Functions

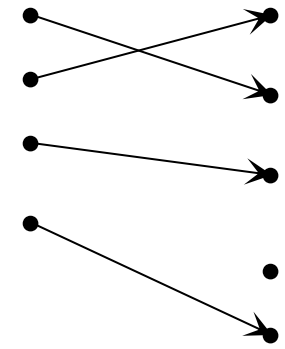
- A function is *one-to-one (1-1)*, or *injective*, or *an injection*, iff every element of its range has *only* 1 pre-image.
  - Formally: given $f{:}A \to B$,
    "x is injective" $:\equiv (\neg\exists x,y{:}\ x \neq y \wedge f(x)=f(y))$.
- Only <u>one</u> element of the domain is mapped <u>to</u> any given <u>one</u> element of the range.
  - Domain & range have same cardinality. What about codomain?



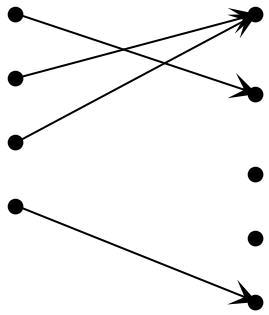
- Memory jogger: Each element of the domain is <u>injected</u> into a different element of the range.
  - Compare "each dose of vaccine is injected into a different patient."

One-to-One Illustration

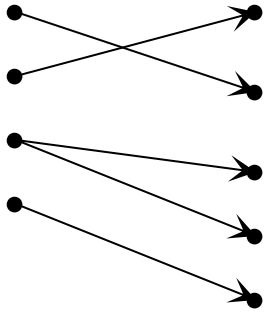
- Bipartite (2-part) graph representations of functions that are (or not) one-to-one:



One-to-one



Not one-to-one



Not even a function!

Sufficient Conditions for 1-1ness

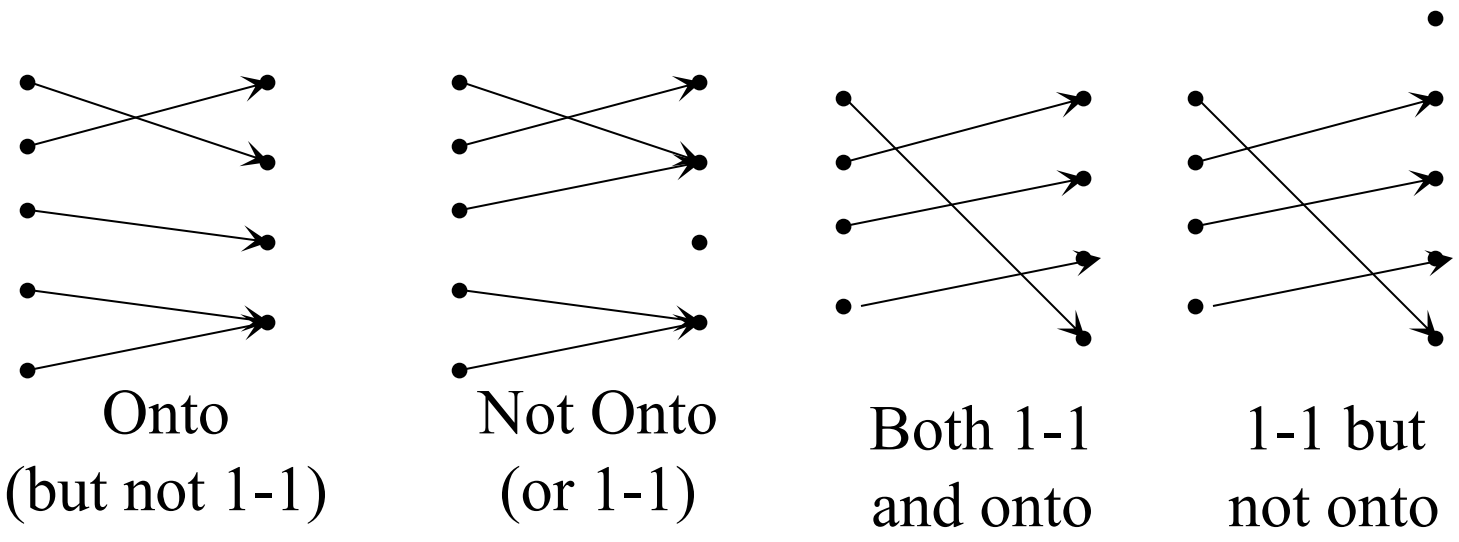
- For functions *f* over numbers, we say:
  - *f* is *strictly* (or *monotonically*) *increasing* iff $x>y \rightarrow f(x)>f(y)$ for all *x,y* in domain;
  - *f* is *strictly* (or *monotonically*) *decreasing* iff $x>y \rightarrow f(x)<f(y)$ for all *x,y* in domain;
- If *f* is either strictly increasing or strictly decreasing, then *f* is one-to-one. *E.g.* x^3
  - *Converse is not necessarily true. E.g.* $1/x$

Onto (Surjective) Functions

- A function $f:A \to B$ is *onto* or *surjective* or *a surjection* iff its range is equal to its codomain ($\forall b \in B, \exists a \in A: f(a)=b$).
- Think: An *onto* function maps the set *A* onto (over, covering) the *entirety* of the set *B*, not just over a piece of it.
- *E.g.*, for domain & codomain **R**, x^3 is onto, whereas x^2 isn't. (Why not?)

Illustration of Onto

- Some functions that are, or are not, *onto* their codomains:



Showing that *f* is one-to-one or onto

Suppose that $f : A \to B$.

To show that f is injective Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$ with $x \neq y$, then $x = y$.

To show that f is not injective Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

To show that f is surjective Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

To show that f is not surjective Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Bijections

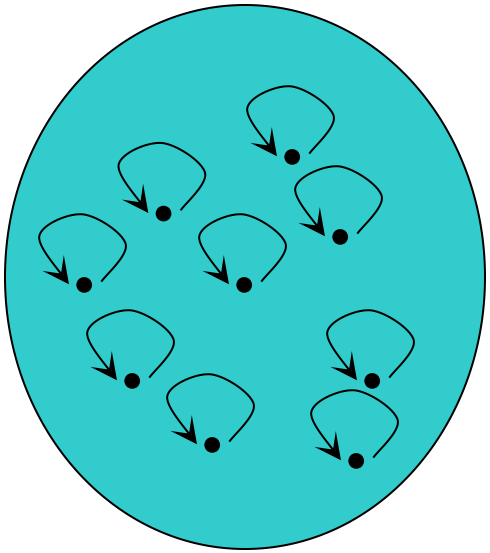
- A function *f* is said to be *a one-to-one correspondence*, or *a bijection*, or *reversible*, or *invertible*, iff it is both one-to-one and onto.
- For bijections $f:A \rightarrow B$, there exists an *inverse of f*, written $f^{-1}:B \rightarrow A$, which is the unique function such that $f^{-1} \circ f = I_A$
  - (where I_A is the identity function on A)

The Identity Function

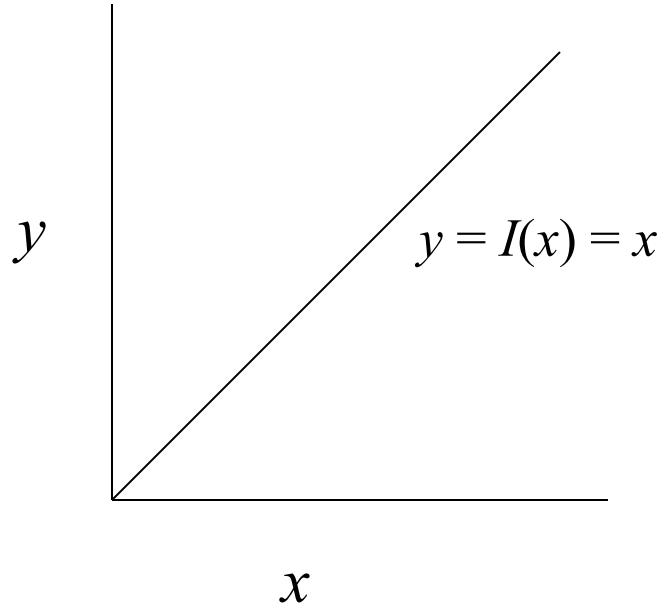
- For any domain A, the *identity function* $I{:}A\to A$ (variously written, I_A, $\mathbf{1}$, $\mathbf{1}_A$) is the unique function such that $\forall a\in A{:}\ I(a)=a$.
- Some identity functions you've seen:
  - $+$ing 0, $\cdot$ing by 1, $\wedge$ing with **T**, $\vee$ing with **F**, $\cup$ing with $\varnothing$, $\cap$ing with U.
- Note that the identity function is always both one-to-one and onto (bijective).

Identity Function Illustrations

- The identity function:



Domain and range



A Cool (Literally) Application

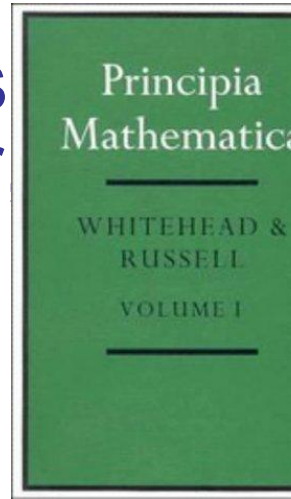
- In a computer, the function mapping the machine's state at clock cycle #*t* to its state at clock cycle #*t*+1 is called the computer's *transition function*.
- If the transition function happens to be reversible (a bijection), then the computer's operation, in theory, requires no energy expenditure.
- The study of practical low-power *reversible computing* techniques based on this idea is an area of research.

Graphs of Functions

- We can represent a function $f{:}A \to B$ as a set of ordered pairs $\{(a,f(a))$ ← The function's *graph*.
- Note that $\forall a$, there is only 1 pair (a,b).
  - Later (ch.9): *relations* loosen this restriction.
- For functions over numbers, we can represent an ordered pair (x,y) as a point on a plane.
  - A function is then drawn as a curve (set of points), with only one y for each x.

Aside About Representations

- It is possible to represent any type of discrete structure (propositions, bit-strings, numbers, sets, ordered pairs, functions) in terms of virtually any of the other structures (or some combination thereof).
- Probably none of these structures is truly more fundamental than the others (whatever that would mean). However strings, logic, and sets are often used as
the foundation for all else. *E.g.* in ➔

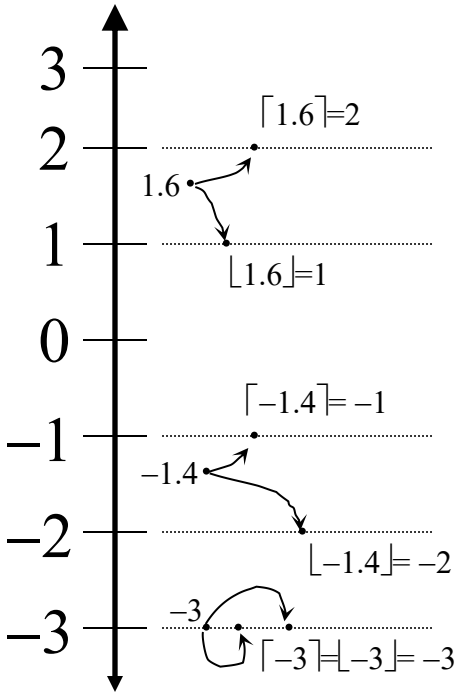


A Couple of Key Functions

- In discrete math, we will frequently use the following two functions over real numbers:
  - The *floor* function $\lfloor \cdot \rfloor : \mathbf{R} \to \mathbf{Z}$, where $\lfloor x \rfloor$ ("floor of x") means the largest (most positive) integer $\leq x$. *I.e.*, $\lfloor x \rfloor :\equiv \max(\{i \in \mathbf{Z} | i \leq x\})$.
  - The *ceiling* function $\lceil \cdot \rceil : \mathbf{R} \to \mathbf{Z}$, where $\lceil x \rceil$ ("ceiling of x") means the smallest (most negative) integer $\geq x$. $\lceil x \rceil :\equiv \min(\{i \in \mathbf{Z} | i \geq x\})$

Visualizing Floor & Ceiling

- Real numbers "fall to their floor" or "rise to their ceiling."
- Note that if $x \notin \mathbf{Z}$, $\lfloor -x \rfloor \neq -\lfloor x \rfloor$ & $\lceil -x \rceil \neq -\lceil x \rceil$
- Note that if $x \in \mathbf{Z}$, $\lfloor x \rfloor = \lceil x \rceil = x$.

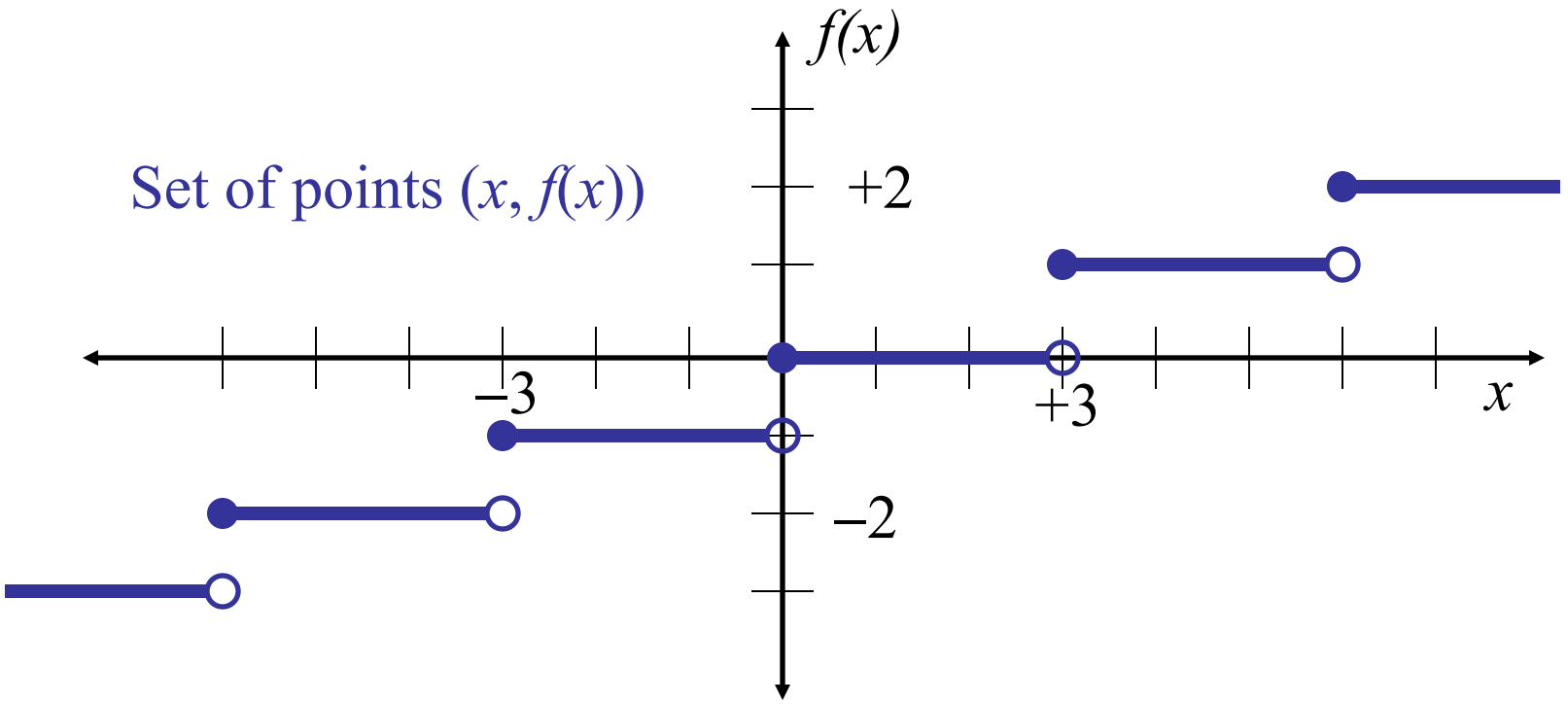


Plots with floor/ceiling

- Note that for $f(x)=\lfloor x \rfloor$, the graph of f includes the point $(a, 0)$ for all values of a such that $a \geq 0$ and $a < 1$, but not for the value $a=1$.
- We say that the set of points $(a,0)$ that is in f does not include its *limit* or *boundary* point $(a,1)$.
  - Sets that do not include all of their limit points are generally called *open sets*.
- In a plot, we draw a limit point of a curve using an open dot (circle) if the limit point is not on the curve, and with a closed (solid) dot if it is on the curve.

Plots with floor/ceiling: Example

- Plot of graph of function $f(x) = \lfloor x/3 \rfloor$:



Floor and Ceiling Functions

TABLE 1 Useful Properties of the Floor and Ceiling Functions.
(n is an integer, x is a real number)

(1a)	$\lfloor x \rfloor = n$ if and only if $n \leq x < n + 1$
(1b)	$\lceil x \rceil = n$ if and only if $n - 1 < x \leq n$
(1c)	$\lfloor x \rfloor = n$ if and only if $x - 1 < n \leq x$
(1d)	$\lceil x \rceil = n$ if and only if $x \leq n < x + 1$
(2)	$x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$
(3a)	$\lfloor -x \rfloor = -\lceil x \rceil$
(3b)	$\lceil -x \rceil = -\lfloor x \rfloor$
(4a)	$\lfloor x + n \rfloor = \lfloor x \rfloor + n$
(4b)	$\lceil x + n \rceil = \lceil x \rceil + n$

Factorial Function

Definition: f: **N** → **Z**$^+$, denoted by $f(n) = n!$ is the product of the first n positive integers when n is a nonnegative integer.

$f(n) = 1 \cdot 2 \cdots (n-1) \cdot n, \qquad f(0) = 0! = 1$

Examples:

$f(1) = 1! = 1$

$f(2) = 2! = 1 \cdot 2 = 2$

$f(6) = 6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720$

$f(20) = 2{,}432{,}902{,}008{,}176{,}640{,}000.$

Stirling's Formula:

$$n! \sim \sqrt{2\pi n}(n/e)^n$$

$$f(n) \sim g(n) \doteq lim_{n\to\infty} f(n)/g(n) = 1$$

Partial Functions

Definition: A *partial function f* from a set *A* to a set *B* is an assignment to each element *a* in a subset of *A*, called the *domain of definition* of *f*, of a unique element *b* in *B*.

- The sets *A* and *B* are called the *domain* and *codomain* of *f*, respectively.
- We day that *f* is *undefined* for elements in *A* that are not in the domain of definition of *f*.
- When the domain of definition of *f* equals *A*, we say that *f* is a *total function*.

Example: *f*: **N** → **R** where $f(n) = \sqrt{n}$ is a partial function from **Z** to **R** where the domain of definition is the set of nonnegative integers. Note that *f* is undefined for negative integers.

Review of § 2.3 (Functions)

- Function variables f, g, h, …
- Notations: $f{:}A \to B$, $f(a)$, $f(A)$.
- Terms: image, preimage, domain, codomain, range, one-to-one, onto, strictly (in/de)creasing, bijective, inverse, composition.
- Function unary operator f^{-1}, binary operators $+$, $-$, *etc.*, and $\bigcirc$.
- The $\mathbf{R} \to \mathbf{Z}$ functions $\lfloor x \rfloor$ and $\lceil x \rceil$.

Sequences

Rosen 8th ed., § 2.4

§ 2.4: Sequences, Strings, & Summations

- A *sequence* or *series* is just like an ordered *n*-tuple, except:
  - Each element in the series has an associated *index* number.
  - A sequence or series may be infinite.
- A *string* is a sequence of *symbols* from some finite *alphabet*.
- A *summation* is a compact notation for the sum of all terms in a (possibly infinite) series.

Sequences

- A *sequence* or *series* $\{a_n\}$ is identified with a *generating function* $f:S\to A$ for some subset $S\subseteq\mathbf{N}$ and for some set A.
  - Often we have $S=\mathbf{N}$ or $S=\mathbf{Z}^+=\mathbf{N}-\{0\}$.
  - Sequences may also be generalized to *indexed sets*, in which the set S does *not* have to be a subset of **N**.
    - For general indexed sets, S may not even be a set of numbers at all.
- If f is a generating function for a series $\{a_n\}$, then for $n\in S$, the symbol a_n denotes $f(n)$, also called *term n* of the sequence.
  - The *index* of a_n is n. (Or, often i is used.)
- A series is sometimes denoted by listing its first and/or last few elements, and using ellipsis (…) notation.
  - *E.g.*, "$\{a_n\} = 0, 1, 4, 9, 16, 25, \ldots$" is taken to mean $\forall n\in\mathbf{N}, a_n = n^2$.

Recurrence Relations

Definition: A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

- A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation.
- The *initial conditions* for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Questions about Recurrence Relations

Example 1: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1,2,3,4,\ldots$ and suppose that $a_0 = 2$. What are a_1, a_2 and a_3?

[Here $a_0 = 2$ is the initial condition.]

Solution: We see from the recurrence relation that

$a_1 = a_0 + 3 = 2 + 3 = 5$

$a_2 = 5 + 3 = 8$

$a_3 = 8 + 3 = 11$

Questions about Recurrence Relations

Example 2: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2,3,4,....$ and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3?

[Here the initial conditions are $a_0 = 3$ and $a_1 = 5$.]

Solution: We see from the recurrence relation that

$$a_2 = a_1 - a_0 = 5 - 3 = 2$$

$$a_3 = a_2 - a_1 = 2 - 5 = -3$$

Fibonacci Sequence

Definition: Define the *Fibonacci sequence*, $f_0, f_1, f_2, \ldots,$ by:

- Initial Conditions: $f_0 = 0$, $f_1 = 1$
- Recurrence Relation: $f_n = f_{n-1} + f_{n-2}$

Example: Find f_2, f_3, f_4, f_5 and f_6 .

Answer:

$$f_2 = f_1 + f_0 = 1 + 0 = 1,$$
$$f_3 = f_2 + f_1 = 1 + 1 = 2,$$
$$f_4 = f_3 + f_2 = 2 + 1 = 3,$$
$$f_5 = f_4 + f_3 = 3 + 2 = 5,$$
$$f_6 = f_5 + f_4 = 5 + 3 = 8.$$

Solving Recurrence Relations

- Finding a formula for the *n*th term of the sequence generated by a recurrence relation is called *solving the recurrence relation*.
- Such a formula is called a ***closed formula***.
- Various methods for solving recurrence relations will be covered in Chapter 8 where recurrence relations will be studied in greater depth.
- Here we illustrate by example the method of iteration in which we need to guess the formula. The guess can be proved correct by the method of induction (Chapter 5).

Iterative Solution Example

Method 1: Working upward, forward substitution

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2,3,4,....$ and suppose that $a_1 = 2$.

$a_2 = 2 + 3$

$a_3 = (2 + 3) + 3 = 2 + 3 \cdot 2$

$a_4 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$

.

.

.

$a_n = a_{n-1} + 3 = (2 + 3 \cdot (n - 2)) + 3 = 2 + 3(n - 1)$

Iterative Solution Example

Method 2: Working downward, backward substitution

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2,3,4,....$ and suppose that $a_1 = 2$.

$$a_n = a_{n-1} + 3$$
$$= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2$$
$$= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3$$
$$\vdots$$
$$= a_2 + 3(n-2) = (a_1 + 3) + 3(n-2) = 2 + 3(n-1)$$

Financial Application

Example: Suppose that a person deposits $\$10{,}000.00$ in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after 30 years?

Let P_n denote the amount in the account after 30 years. P_n satisfies the following recurrence relation:

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11)\,P_{n-1}$$

with the initial condition $P_0 = 10{,}000$

Continued on next slide →

Financial Application

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11)\,P_{n-1}$$

with the initial condition $P_0 = 10{,}000$

Solution: Forward Substitution

$P_1 = (1.11)P_0$

$P_2 = (1.11)P_1 = (1.11)^2P_0$

$P_3 = (1.11)P_2 = (1.11)^3P_0$

:

$P_n = (1.11)P_{n-1} = (1.11)^nP_0 = (1.11)^n\ 10{,}000$

$P_n = (1.11)^n\ 10{,}000$ (Can prove by induction, covered in Chapter 5)

$P_{30} = (1.11)^{30}\ 10{,}000 = \$228{,}992.97$

Special Integer Sequences

- Given a few terms of a sequence, try to identify the sequence. Conjecture a formula, recurrence relation, or some other rule.
- Some questions to ask?
  - Are there repeated terms of the same value?
  - Can you obtain a term from the previous term by adding an amount or multiplying by an amount?
  - Can you obtain a term by combining the previous terms in some way?
  - Are they cycles among the terms?
  - Do the terms match those of a well known sequence?

Questions on Special Integer Sequences

Example 1: Find formulae for the sequences with the following first five terms: 1, ½, ¼, 1/8, 1/16

Solution: Note that the denominators are powers of 2. The sequence with $a_n = 1/2^n$ is a possible match. This is a geometric progression with $a = 1$ and $r = ½$.

Example 2: Consider 1,3,5,7,9

Solution: Note that each term is obtained by adding 2 to the previous term. A possible formula is $a_n = 2n + 1$. This is an arithmetic progression with $a = 1$ and $d = 2$.

Example 3: 1, -1, 1, -1,1

Solution: The terms alternate between 1 and -1. A possible sequence is $a_n = (-1)^n$. This is a geometric progression with $a = 1$ and $r = -1$.

Useful Sequences

TABLE 1 **Some Useful Sequences.**

nth Term	*First 10 Terms*
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, . . .
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, . . .
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, . . .
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, . . .
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, . . .
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, . . .
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, . . .

Guessing Sequences

Example: Conjecture a simple formula for a_n if the first 10 terms of the sequence $\{a_n\}$ are $1, 7, 25, 79, 241, 727, 2185, 6559, 19681, 59047$.

Solution: Note the ratio of each term to the previous approximates 3. So now compare with the sequence 3^n. We notice that the nth term is 2 less than the corresponding power of 3. So a good conjecture is that $a_n = 3^n - 2$.

Integer Sequences

- Integer sequences appear in a wide range of contexts. Later we will see the sequence of prime numbers (Chapter 4), the number of ways to order *n* discrete objects (Chapter 6), the number of moves needed to solve the Tower of Hanoi puzzle with *n* disks (Chapter 8), and the number of rabbits on an island after *n* months (Chapter 8).
- Integer sequences are useful in many fields such as biology, engineering, chemistry and physics.
- On-Line Encyclopedia of Integer Sequences (OESIS) contains over 200,000 sequences. Began by Neil Stone in the 1960s (printed form). Now found at http://oeis.org/Spuzzle.html

Integer Sequences

- Here are three interesting sequences to try from the OESIS site. To solve each puzzle, find a rule that determines the terms of the sequence.
- Guess the rules for forming for the following sequences:
  - 2, 3, 3, 5, 10, 13, 39, 43, 172, 177, ...
    - Hint: Think of adding and multiplying by numbers to generate this sequence.
  - 0, 0, 0, 0, 4, 9, 5, 1, 1, 0, 55, ...
    - Hint: Think of the English names for the numbers representing the position in the sequence and the Roman Numerals for the same number.
  - 2, 4, 6, 30, 32, 34, 36, 40, 42, 44, 46, ...
    - Hint: Think of the English names for numbers, and whether or not they have the letter 'e.'

- The answers and many more can be found at http://oeis.org/Spuzzle.html

Sequence Examples

- Some authors write "the sequence a_1, a_2, …" instead of $\{a_n\}$, to ensure that the set of indices is clear.
  - Be careful: Our book often leaves the indices ambiguous.
- An example of an infinite series:
  - Consider the series $\{a_n\} = a_1, a_2, \ldots$, where $(\forall n \geq 1)\ a_n = f(n) = 1/n$.
  - Then, we have $\{a_n\} = 1, 1/2, 1/3, \ldots$

Example with Repetitions

- Like tuples, but unlike sets, a sequence may contain *repeated* instances of an element.
- Consider the sequence $\{b_n\} = b_0, b_1, \ldots$ (note that 0 is an index) where $b_n = (-1)^n$.
  - Thus, $\{b_n\} = 1, -1, 1, -1, \ldots$
    - Note repetitions!
  - This $\{b_n\}$ denotes an infinite sequence of 1's and –1's, *not* the 2-element set $\{1, -1\}$.

Recognizing Sequences

- Sometimes, you're given the first few terms of a sequence,
  - and you are asked to find the sequence's generating function,
  - or a procedure to enumerate the sequence.
- Examples: What's the next number?
  - 1,2,3,4,… 5 (the 5th smallest number >0)
  - 1,3,5,7,9,… 11 (the 6th smallest odd number >0)
  - 2,3,5,7,11,... 13 (the 6th smallest prime number)
- 0，1，2，5，12，27....

The Trouble with Sequence Recognition

- As you know, these problems are popular on IQ tests, but…
- The problem of finding "the" generating function given just an initial subsequence is *not a mathematically well defined problem.*
  - This is because there are *infinitely* many computable functions that will generate *any* given initial subsequence.
- We implicitly are supposed to find the *simplest* such function (because this one is assumed to be most likely), but,
  - how are we to objectively define the *simplicity* of a function?
- We might define simplicity as the reciprocal of complexity, but…
  - There are *many* different plausible, competing definitions of complexity, and this is an active research area.
- So, these questions really have *no* objective right answer!
  - Still, we will ask you to answer them anyway… (Because others will too.)

What are Strings, Really?

- This book says "finite sequences of the form $a_1, a_2, \ldots, a_n$ are called *strings*",
  - but *infinite* strings are also discussed sometimes.
- Strings are normally restricted to sequences composed of *symbols* drawn from a finite *alphabet*, and are often indexed from 0 or 1.
  - But these are really arbitrary restrictions also.
- Either way, the *length* of a (finite) string is just its number of terms (or of distinct indices).

Strings, more formally

- Let Σ be a finite set of *symbols*, *i.e.* an *alphabet.*
  - A *string* s over alphabet Σ is any sequence $\{s_i\}$ of symbols, $s_i \in \Sigma$, normally indexed by **N** or **N**–{0}.
- If a, b, c, … are symbols, the string $s = a, b, c, \ldots$ can also be written $abc\ldots$ (*i.e.*, without commas).
- If s is a finite string and t is any string, then the *concatenation of s with t*, written just st,
  - is simply the string consisting of the symbols in s, in sequence, followed by the symbols in t, in sequence.

More Common String Notations

- The length $|s|$ of a finite string s is its number of *positions* (*i.e.*, its number of index values i).
- If s is a finite string and $n \in \mathbf{N}$,
  - Then s^n denotes the concatenation of n copies of s.
- ε or "" denotes the empty string, the string of length 0.
  - This is fairly common, but the book uses λ instead.
- If Σ is an alphabet and $n \in \mathbf{N}$,

  $\Sigma^n :\equiv \{s \mid s \text{ is a string over } \Sigma \text{ of length } n\}$, and

  $\Sigma^* :\equiv \{s \mid s \text{ is a finite string over } \Sigma\}$.

设计和艺术创作与数列

- 调和数列：　调和数列变化特征是：具有女性优雅、细腻、柔美之感，同时还具有一种宁静之美。
- 费波纳齐数列(FIBONACCI)：　费波纳齐为意大利数学家。在自然界中螺纹形贝壳均属费波纳齐数列现象。其特征：这是一种不疾不徐，颇为中庸的数列，增加率具有规则性的平稳，令人感到愉快的、优美的数列。它兼有等差数列与等比数列的优点。
- 等比数列：特征：等比数列在开始时虽然变化不大，但是越往后面，其变化则越加剧烈，这是一个强有力的数列，其变化基于几何数列。美感在于大小可以急剧的增加或减少，造成一种剧烈火暴的感觉。如果能够活用等差数列的特征，则能创造出富于微妙变化的设计和画面。
- 例如：慕尼黑奥林匹克运动会的商标，日本万国博览会的商标等，都是在设计中运用了等比数列的例子，并取得了很好的视觉效果。
- 贝尔数列(BELL)：　特征：它与等差数列相同，美感也在于大小可以急剧的增加或减少之中。造成一种剧烈、火爆且强劲有力的感觉。



Summations

Summation Notation

- Given a series $\{a_n\}$, an integer *lower bound* (or *limit*) $j \geq 0$, and an integer *upper bound* $k \geq j$, then the *summation of* $\{a_n\}$ *from j to k* is written and defined as follows:

$$\sum_{i=j}^{k} a_i :\equiv a_j + a_{j+1} + \ldots + a_k$$

- Here, *i* is called the *index of summation*.

Generalized Summations

- For an infinite series, we may write:

$$\sum_{i=j}^{\infty} a_i \equiv a_j + a_{j+1} + \ldots$$

- To sum a function over all members of a set $X=\{x_1, x_2, \ldots\}$: $\sum_{x \in X} f(x) \equiv f(x_1) + f(x_2) + \ldots$
- Or, if $X=\{x|P(x)\}$, we may just write:

$$\sum_{P(x)} f(x) \equiv f(x_1) + f(x_2) + \ldots$$

Simple Summation Example

$$\sum_{i=2}^{4} i^2 + 1 = (2^2 + 1) + (3^2 + 1) + (4^2 + 1)$$

$$= (4 + 1) + (9 + 1) + (16 + 1)$$

$$= 5 + 10 + 17$$

$$= 32$$

More Summation Examples

- An infinite series with a finite sum:

$$\sum_{i=0}^{\infty} 2^{-i} = 2^0 + 2^{-1} + \ldots = 1 + \tfrac{1}{2} + \tfrac{1}{4} + \ldots = 2$$

- Using a predicate to define a set of elements to sum over:

$$\sum_{\substack{(x \text{ is prime}) \wedge \\ x<10}} x^2 = 2^2 + 3^2 + 5^2 + 7^2 = 4 + 9 + 25 + 49 = 87$$

Summation Manipulations

- Some handy identities for summations:

$$\sum_x cf(x) = c\sum_x f(x)$$ (Distributive law.)

$$\sum_x f(x) + g(x) = \left(\sum_x f(x)\right) + \sum_x g(x)$$ (An application of commut-ativity.)

$$\sum_{i=j}^{k} f(i) = \sum_{i=j+n}^{k+n} f(i-n)$$ (Index shifting.)

More Summation Manipulations

- Other identities that are sometimes useful:

$$\sum_{i=j}^{k} f(i) = \left(\sum_{i=j}^{m} f(i) \right) + \sum_{i=m+1}^{k} f(i) \quad \text{if } j \le m < k$$

(Series splitting.)

$$\sum_{i=0}^{k} f(i) = \sum_{i=0}^{k} f(k-i)$$

(Order reversal.)

$$\sum_{i=0}^{2k} f(i) = \sum_{i=0}^{k} f(2i) + f(2i+1)$$

(Grouping.)

Example: Impress Your Friends

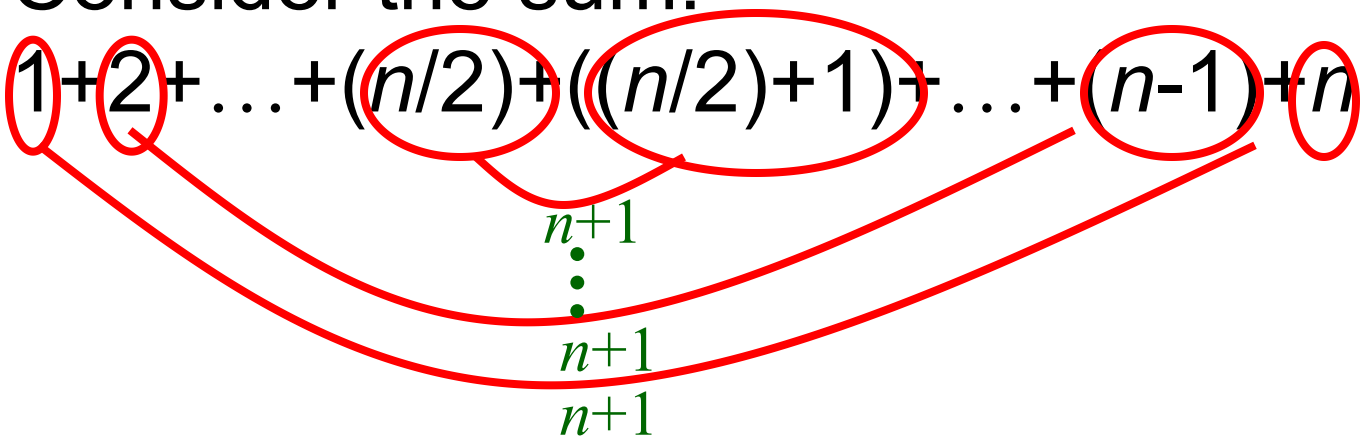
- Boast, "I'm so smart; give me any 2-digit number *n*, and I'll add all the numbers from 1 to *n* in my head in just a few seconds."
- *I.e.*, Evaluate the summation: $\sum_{i=1}^{n} i$
- There is a simple closed-form formula for the result, discovered by Euler at age 12!
  - And frequently rediscovered by many…

Leonhard Euler (1707-1783)



Euler's Trick, Illustrated

- Consider the sum:



- We have $n/2$ pairs of elements, each pair summing to $n+1$, for a total of $(n/2)(n+1)$.

Symbolic Derivation of Trick

For case where n is even…

$$\sum_{i=1}^{n} i = \sum_{i=1}^{2k} i = \left(\sum_{i=1}^{k} i\right) + \sum_{i=k+1}^{n} i = \left(\sum_{i=1}^{k} i\right) + \sum_{i=0}^{n-(k+1)} (i + (k+1))$$

$$= \left(\sum_{i=1}^{k} i\right) + \sum_{i=0}^{n-(k+1)} ((n-(k+1)) - i) + (k+1))$$

$$= \left(\sum_{i=1}^{k} i\right) + \sum_{i=0}^{n-(k+1)} (n-i) = \left(\sum_{i=1}^{k} i\right) + \sum_{i=1}^{n-k} (n-(i-1))$$

$$= \left(\sum_{i=1}^{k} i\right) + \sum_{i=1}^{n-k} (n+1-i) = \left(\sum_{i=1}^{k} i\right) + \sum_{i=1}^{k} (n+1-i) = \ldots$$

Concluding Euler's Derivation

$$\sum_{i=1}^{n} i = \left(\sum_{i=1}^{k} i \right) + \sum_{i=1}^{k} (n+1-i) = \sum_{i=1}^{k} (i+n+1-i)$$

$$= \sum_{i=1}^{k} (n+1) = k(n+1) = \tfrac{n}{2}(n+1)$$

$$= n(n+1)/2$$

- So, you only have to do 1 easy multiplication in your head, then cut in half.
- Also works for odd *n* (prove this at home).

Example: Geometric Progression

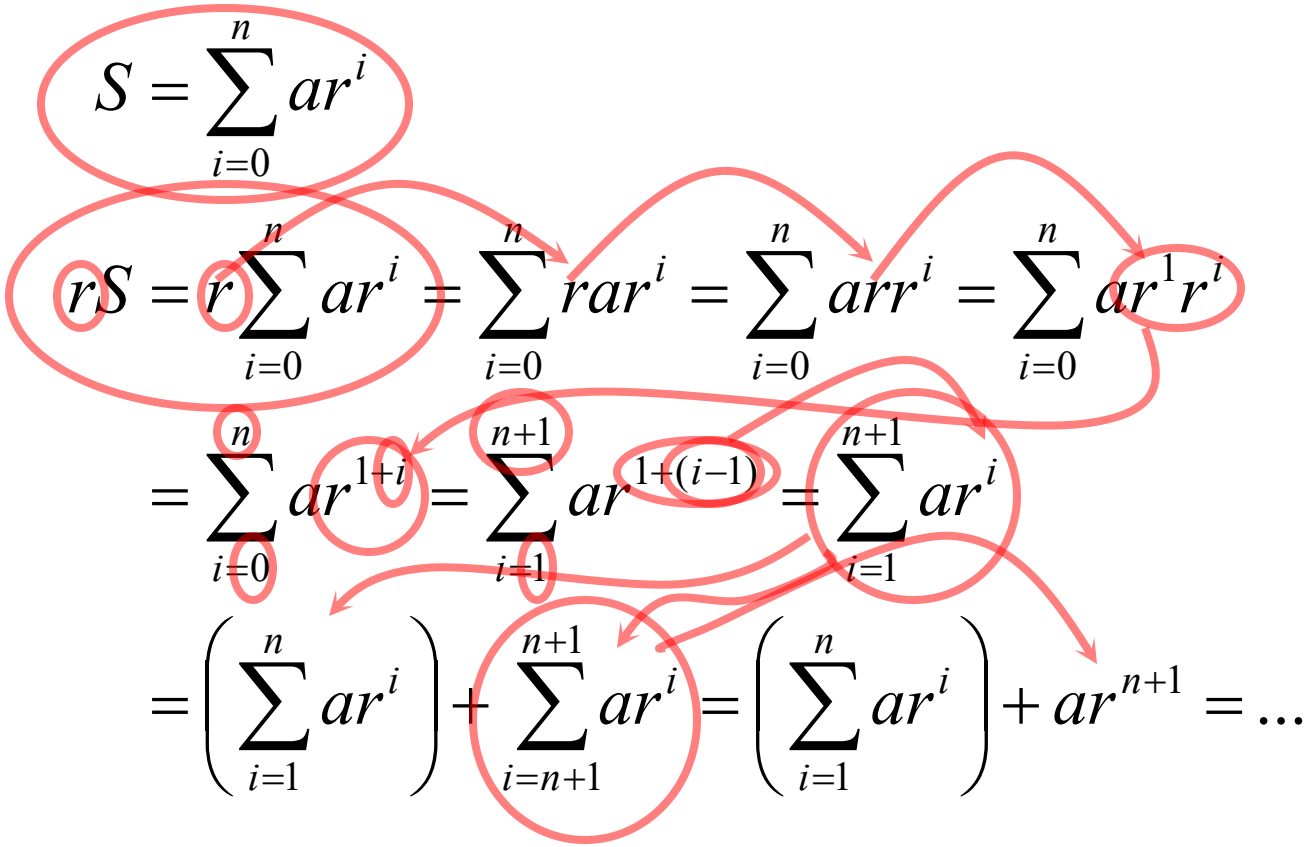
- A *geometric progression* is a series of the form a, ar, ar^2, ar^3, …, ar^k, where $a,r \in \mathbf{R}$.
- The sum of such a series is given by:

$$S = \sum_{i=0}^{k} ar^i$$

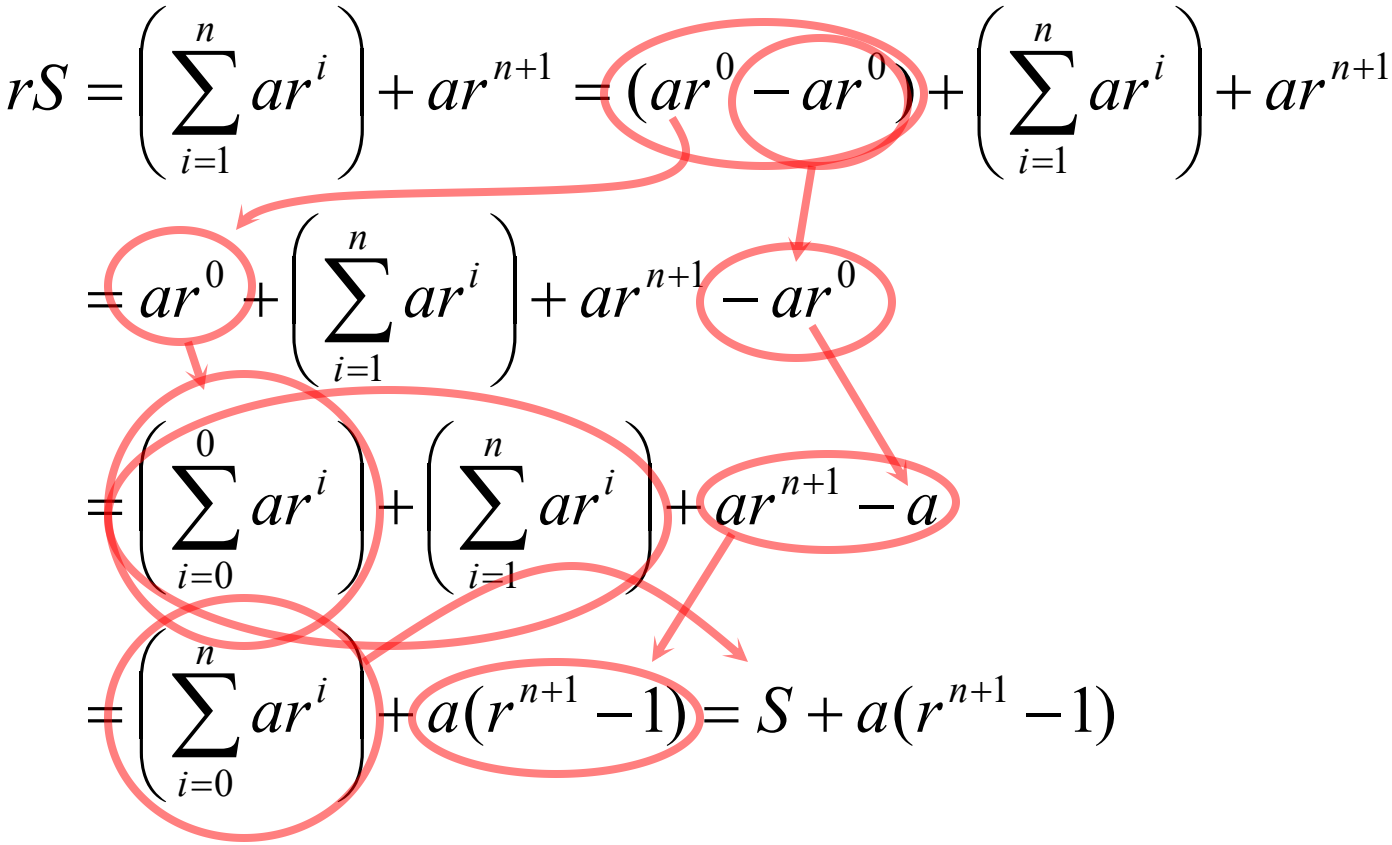
- We can reduce this to *closed form* via clever manipulation of summations...

Geometric Sum Derivation

- Here we go...



Derivation example cont...



Concluding long derivation...

$$rS = S + a(r^{n+1} - 1)$$

$$rS - S = a(r^{n+1} - 1)$$

$$S(r-1) = a(r^{n+1} - 1)$$

$$S = a\left(\frac{r^{n+1} - 1}{r - 1}\right) \qquad \text{when } r \neq 1$$

$$\text{When } r = 1, S = \sum_{i=0}^{n} ar^i = \sum_{i=0}^{n} a1^i = \sum_{i=0}^{n} a \cdot 1 = (n+1)a$$

Nested Summations

- These have the meaning you'd expect.

$$\sum_{i=1}^{4}\sum_{j=1}^{3} ij = \sum_{i=1}^{4}\left(\sum_{j=1}^{3} ij\right) = \sum_{i=1}^{4} i\left(\sum_{j=1}^{3} j\right) = \sum_{i=1}^{4} i(1+2+3)$$

$$= \sum_{i=1}^{4} 6i = 6\sum_{i=1}^{4} i = 6(1+2+3+4)$$

$$= 6 \cdot 10 = 60$$

- Note issues of free vs. bound variables, just like in quantified expressions, integrals, etc.

Some Shortcut Expressions

$$\sum_{k=0}^{n} ar^k = a(r^{n+1}-1)/(r-1), r \neq 1$$ Geometric series.

$$\sum_{k=1}^{n} k = n(n+1)/2$$ Euler's trick.

$$\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$$ Quadratic series.

$$\sum_{k=1}^{n} k^3 = n^2(n+1)^2/4$$ Cubic series.

Using the Shortcuts

- Example: Evaluate $\sum_{k=50}^{100} k^2$.
  - Use series splitting.
  - Solve for desired summation.
  - Apply quadratic series rule.
  - Evaluate.

$$\sum_{k=1}^{100} k^2 = \left(\sum_{k=1}^{49} k^2\right) + \sum_{k=50}^{100} k^2$$

$$\sum_{k=50}^{100} k^2 = \left(\sum_{k=1}^{100} k^2\right) - \sum_{k=1}^{49} k^2$$

$$= \frac{100 \cdot 101 \cdot 201}{6} - \frac{49 \cdot 50 \cdot 99}{6}$$

$$= 338{,}350 - 40{,}425$$

$$= 297{,}925.$$

Some Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.

Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x \rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, \lvert x \rvert < 1$	$\frac{1}{(1-x)^2}$

Geometric Series: We just proved this.

Later we will prove some of these by induction.

Proof in text (requires calculus)

Summations: Conclusion

- You need to know:
  - How to read, write & evaluate summation expressions like:

$$\sum_{i=j}^{k} a_i \qquad \sum_{i=j}^{\infty} a_i \qquad \sum_{x \in X} f(x) \qquad \sum_{P(x)} f(x)$$

  - Summation manipulation laws we covered.
  - Shortcut closed-form formulas, & how to use them.

2.5 Cardinality & Infinite Sets

Infinite Cardinalities

- Using what we learned about *functions* in § 2.3, it's possible to formally define cardinality even for infinite sets.
- We can also show that infinite sets come in different *sizes* of infinite!
- This also gives us some interesting proof examples.

Cardinality: Formal Definition

- For any two (possibly infinite) sets A and B, we say that A and B *have the same cardinality* (written $|A|=|B|$) iff there exists a bijection (bijective function) from A to B.
- When A and B are finite, it is easy to see that such a function exists iff A and B have the same number of elements $n \in \mathbf{N}$.

Countable versus Uncountable

- For any set S, if S is finite or if $|S|=|\mathbf{N}|$, we say S is *countable*. Else, S is *uncountable.*
- Intuition behind "**countable**:" we can *enumerate* (sequentially list) elements of S in such a way that *any* individual element of S will eventually be *counted* in the enumeration. Examples: $\mathbf{N}$, $\mathbf{Z}$.
- **Uncountable** means: *No* series of elements of S (even an infinite series) can include all of S's elements. Examples: $\mathbf{R}$, $\mathbf{R}^2$, $P(\mathbf{N})$

Countable Sets: Examples

- **Theorem:** The set **Z** is countable.
  - **Proof:** Consider f:**Z**$\rightarrow$**N** where $f(i)=2i$ for $i\geq 0$ and $f(i) = -2i-1$ for $i<0$. Note f is bijective.
- **Theorem:** The set of all ordered pairs of natural numbers (n,m) is countable.
  - Consider listing the pairs in order by their sum $s=n+m$, then by n. Every pair appears once in this series; the generating function is bijective.

The Positive Rational Numbers are Countable

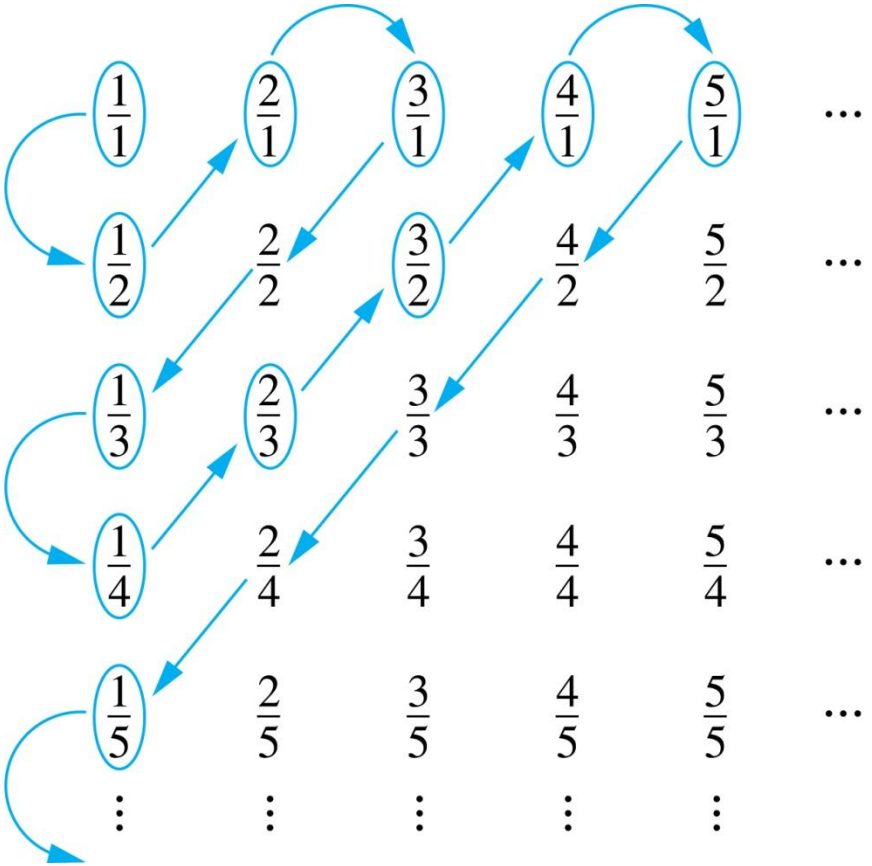
First row $q = 1$.
Second row $q = 2$.
etc.

Constructing the List

First list p/q with $p + q = 2$.
Next list p/q with $p + q = 3$

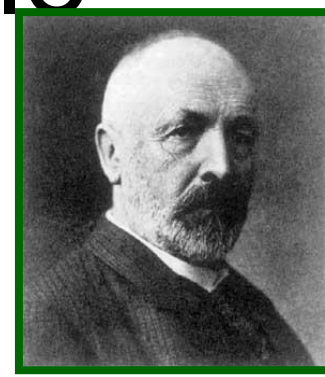
And so on.

Terms not circled are not listed because they repeat previously listed terms



1, ½, 2, 3, 1/3,1/4, 2/3, ...

Uncountable Sets: Example



- **Theorem:** The open interval of reals $[0,1) :\equiv \{r \in \mathbf{R} \mid 0 \leq r < 1\}$ is uncountable.
- **Proof** by *diagonalization*: (Cantor, 1891)
  - Assume there is a series $\{r_i\} = r_1, r_2, \ldots$ containing *all* elements $r \in [0,1)$.
  - Consider listing the elements of $\{r_i\}$ in decimal notation (although any base will do) in order of increasing index: ... *(continued on next slide)*

Georg Cantor
1845-1918

Uncountability of Reals, cont'd

A postulated enumeration of the reals:

$r_1 = 0.d_{1,1}\, d_{1,2}\, d_{1,3}\, d_{1,4}\, d_{1,5}\, d_{1,6}\, d_{1,7}\, d_{1,8}\ldots$

$r_2 = 0.d_{2,1}\, d_{2,2}\, d_{2,3}\, d_{2,4}\, d_{2,5}\, d_{2,6}\, d_{2,7}\, d_{2,8}\ldots$

$r_3 = 0.d_{3,1}\, d_{3,2}\, d_{3,3}\, d_{3,4}\, d_{3,5}\, d_{3,6}\, d_{3,7}\, d_{3,8}\ldots$

$r_4 = 0.d_{4,1}\, d_{4,2}\, d_{4,3}\, d_{4,4}\, d_{4,5}\, d_{4,6}\, d_{4,7}\, d_{4,8}\ldots$

.

.

.

Now, consider a real number generated by taking all the digits $d_{i,i}$ that lie along the *diagonal* in this figure and replacing them with *different* digits.

That real doesn't appear in the list!

Uncountability of Reals, fin.

- *E.g.*, a postulated enumeration of the reals:

  r_1 = 0.301948571...
  r_2 = 0.103918481...
  r_3 = 0.039194193...
  r_4 = 0.918237461...

- OK, now let's add 1 to each of the diagonal digits (mod 10), that is changing 9's to 0.
  - 0.4103… can't be on the list anywhere!

Transfinite Numbers超限数

- The cardinalities of infinite sets are not natural numbers, but are special objects called *transfinite* cardinal numbers.
- The cardinality of the natural numbers, $\aleph_0 :\equiv |\mathbf{N}|$, is the *first transfinite cardinal* number. (There are none smaller.)
- The *continuum hypothesis* claims that $|\mathbf{R}| = \aleph_1$, the *second transfinite cardinal.*



代数数

形如$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$（$a_n \neq 0$，n为正整数）的整系数（$a_k$为整数，$0 \leq k \leq n$）多项式方程的根x则叫做“代数数”。

代数数可以定义为“有理系数多项式的复根”或“整系数多项式的复根”。第一个定义可以具体描述为：

设 z 为复数。如果存在正整数 n，以及 (n + 1) 个有理数$q_0, q_1, \cdots, q_n$，并且$q_n \neq 0$，使得：

$$q_n z^n + \cdots + q_1 z + q_0 = 0$$

则称 z 是一个代数数。

这个定义中，由于$q_n z^n \cdots + q_1 z + q_0 = 0$可以推出$a_n z^n + \cdots + a_1 z + a_0 = 0$，其中整数$a_0, a_1, \cdots, a_n$分别等于$Mq_0, Mq_1, \cdots, Mq_n$，M 是 (n + 1) 个有理数$q_0, q_1, \cdots, q_n$分母的最小公倍数。所以“存在有理系数多项式使得z是其复根”可以推出“存在整系数多项式使得z是其复根”。另一方面，由于整数集合是有理数集合的子集，所以“存在整系数多项式使得z是其复根”也可以推出“存在有理系数多项式使得 z 是其复根”。这说明两个定义是等价的。

代数数集包含了有理数集。然而，代数数集并不包含全部实数。代数数集是一个可数集，即所有代数数能与全体自然数建立一一对应，而实数集是不可数的无穷集，因此，一定存在不是代数数的实数。现已证明 π和e这些无理数不是代数数，但不是所有的无理数都不是代数数。不是代数数的数称为超越数。由此可见，就实数集而言，实数既可按有理数和无理数分为两类，又可按实代数数和实超越数分为两类。实代数数集是有理数集的自然扩充。

Countable vs. Uncountable

- You should:
  - Know how to define “same cardinality” in the case of infinite sets.
  - Know the definitions of *countable* and *uncountable*.
  - Know how to prove (at least in easy cases) that sets are either countable or uncountable.

作业

- § 2.3　8, 20, 24, 33, 66
- § 2.4 – 16,26, 30, 34
- § 2.5 – 2，10

§ 2.6 Matrices

- A *matrix* is a rectangular array of objects (usually numbers).
- An $m \times n$ ("*m* by *n*") matrix has exactly *m* horizontal rows, and *n* vertical columns.
- Plural of matrix = *matrices* (say MAY-trih-sees)

$$\begin{bmatrix} 2 & 3 \\ 5 & -1 \\ 7 & 0 \end{bmatrix}$$ a 3×2 matrix

- An $n \times n$ matrix is called a *square* matrix, whose *order* or *rank* is *n*.

Applications of Matrices

Tons of applications, including:

- Solving systems of linear equations
- Computer Graphics, Image Processing
- Models within many areas of Computational Science & Engineering
- Quantum Mechanics, Quantum Computing
- Many, many more…

Matrix Equality

- Two matrices **A** and **B** are considered equal iff they have the same number of rows, the same number of columns, and all their corresponding elements are equal.

$$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} \neq \begin{bmatrix} 3 & 2 & 0 \\ -1 & 6 & 0 \end{bmatrix}$$

Row and Column Order

- The rows in a matrix are usually indexed 1 to *m* from top to bottom. The columns are usually indexed 1 to *n* from left to right. Elements are indexed by row, then column.

$$\mathbf{A} = [a_{i,j}] = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

Matrices as Functions

- An $m \times n$ matrix $\mathbf{A} = [a_{i,j}]$ of members of a set S can be encoded as a partial function

$$f_{\mathbf{A}}: \mathbf{N} \times \mathbf{N} \rightarrow S,$$

  such that for $i<m$, $j<n$, $f_{\mathbf{A}}(i, j) = a_{i,j}$.
- By extending the domain over which $f_{\mathbf{A}}$ is defined, various types of infinite and/or multidimensional matrices can be obtained.

Matrix Sums

- The *sum* **A**+**B** of two matrices **A**, **B** (which **must** have the same number of rows, and the same number of columns) is the matrix (also with the same shape) given by adding corresponding elements of **A** and **B**.

$$\mathbf{A}+\mathbf{B} = [a_{i,j}+b_{i,j}]$$

$$\begin{bmatrix} 2 & 6 \\ 0 & -8 \end{bmatrix} + \begin{bmatrix} 9 & 3 \\ -11 & 3 \end{bmatrix} = \begin{bmatrix} 11 & 9 \\ -11 & -5 \end{bmatrix}$$

Matrix Products

- For an $m \times k$ matrix **A** and a $k \times n$ matrix **B**, the *product* **AB** is the $m \times n$ matrix:

$$\mathbf{AB} = \mathbf{C} = [c_{i,j}] \equiv \left[\sum_{\ell=1}^{k} a_{i,\ell} b_{\ell,j} \right]$$

- *I.e.*, the element of **AB** indexed (*i*,*j*) is given by the vector *dot product* of the *i*th row of **A** and the *j*th column of **B** (considered as vectors).
- **Note:** Matrix multiplication is not commutative!

Illustration of Matrix Multiplication

- The Product of **A** = [a_{ij}] and **B** = [b_{ij}]

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ . & . & & . \\ . & . & & . \\ a_{i1} & a_{i2} & \dots & a_{ik} \\ . & . & & . \\ . & . & & . \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & a_{12} & \dots & b_{1j} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2j} & \dots & b_{2n} \\ . & . & & . & & \\ . & . & & . & & \\ b_{k1} & b_{k2} & \dots & b_{kj} & \dots & b_{kn} \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ . & . & & . \\ . & . & c_{ij} & . \\ . & . & & . \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}$$

Matrix Product Example

- An example matrix multiplication to practice in class:

$$\begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 & 1 & 0 \\ 2 & 0 & -2 & 0 \\ 1 & 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -5 & -1 \\ 3 & -2 & 11 & 3 \end{bmatrix}$$

Identity Matrices

- The *identity matrix of order n*, $\mathbf{I}_n$, is the rank-*n* square matrix with 1's along the upper-left to lower-right diagonal, and 0's everywhere else.

$$\mathbf{I}_n = [\delta_{ij}] = \left[\begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases} \right] = \underbrace{\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}}_{n} \; n$$

$\forall 1 \leq i,j \leq n$

Kronecker Delta

Review: Matrices, so far

Matrix sums and products:

$$\mathbf{A}+\mathbf{B} = [a_{i,j}+b_{i,j}]$$

$$\mathbf{AB} = \mathbf{C} = [c_{i,j}] \equiv \left[\sum_{\ell=1}^{k} a_{i,\ell} b_{\ell,j} \right]$$

Identity matrix of order *n*:

$\mathbf{I}_n = [\delta_{ij}]$, where $\delta_{ij}=1$ if $i=j$ and $\delta_{ij}=0$ if $i \neq j$.

Matrix Inverses

- For some (but not all) square matrices $\mathbf{A}$, there exists a unique multiplicative *inverse* $\mathbf{A}^{-1}$ of $\mathbf{A}$, a matrix such that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$.
- If the inverse exists, it is unique, and $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1}$.
- We won't go into the algorithms for matrix inversion...

Matrix Multiplication Algorithm

procedure *matmul*(matrices **A**: $m \times k$, **B**: $k \times n$)

for i := 1 **to** m — $\Theta(m)\cdot$

  for j := 1 **to** n **begin** — $\Theta(n)\cdot($

    c_{ij} := 0 — $\Theta(1)+$

    for q := 1 **to** k — $\Theta(k)\cdot$

      $c_{ij} := c_{ij} + a_{iq}b_{qj}$ — $\Theta(1))$

  end {**C**=[c_{ij}] is the product of **A** and **B**}

What's the Θ of its time complexity?

Answer: $\Theta(mnk)$

Powers of Matrices

If **A** is an $n \times n$ square matrix and $p \geq 0$, then:

- $\mathbf{A}^p :\equiv \underbrace{\mathbf{AAA}\cdots\mathbf{A}}_{p \text{ times}}$ (and $\mathbf{A}^0 :\equiv \mathbf{I}_n$)

- Example:

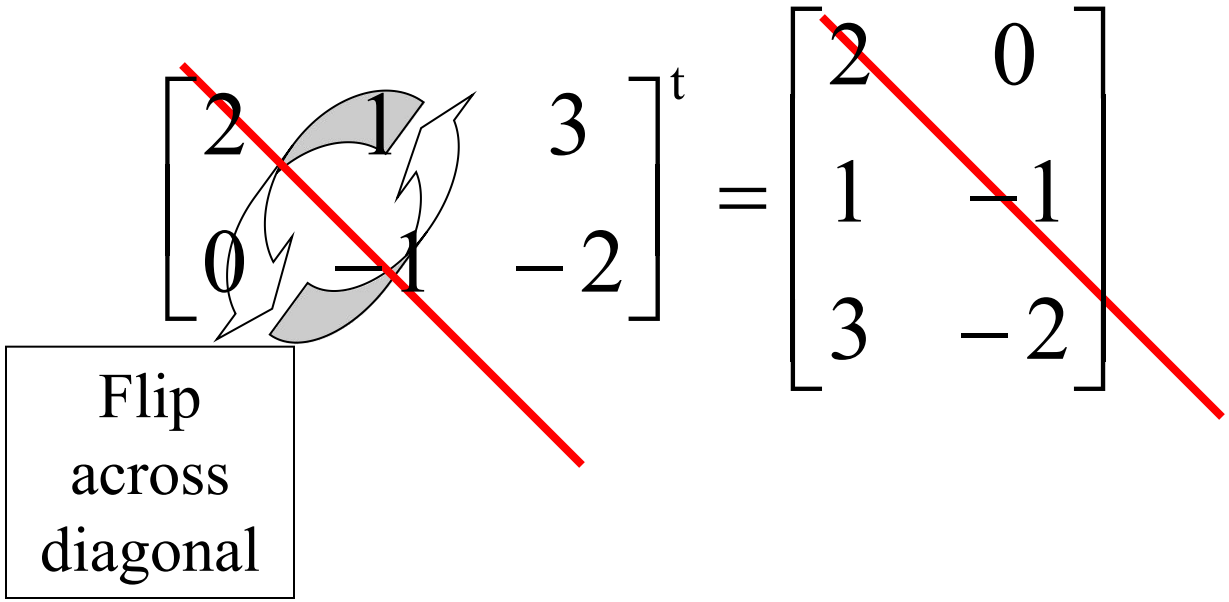
$$\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}^3 = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 3 & 2 \\ -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix}$$

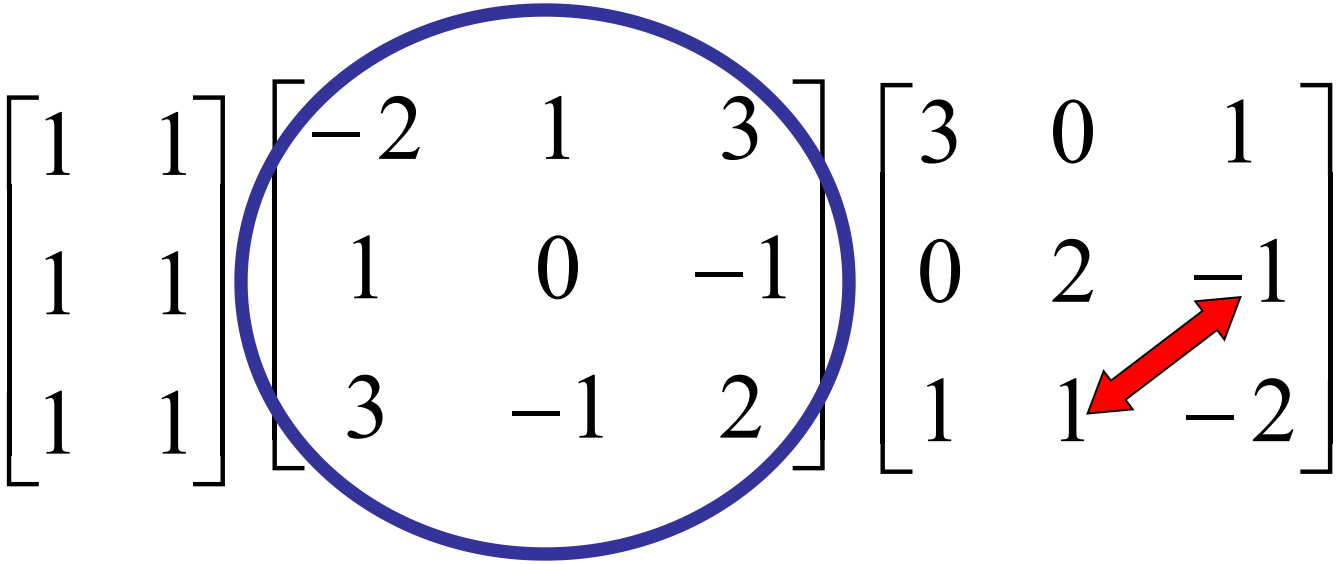
Matrix Transposition

- If $\mathbf{A}=[a_{ij}]$ is an $m\times n$ matrix, the *transpose* of **A** (often written $\mathbf{A}^t$ or $\mathbf{A}^T$) is the $n\times m$ matrix given by $\mathbf{A}^t = \mathbf{B} = [b_{ij}] = [a_{ji}]$ $(1\le i\le n, 1\le j\le m)$



Symmetric Matrices

- A square matrix **A** is *symmetric* iff $\mathbf{A}=\mathbf{A}^t$.
  I.e., $\forall i,j \leq n: a_{ij} = a_{ji}$.
- Which of the below matrices is symmetric?



Zero-One Matrices

- Useful for representing other structures.
  - *E.g.*, relations, directed graphs (later in this course)
- All elements of a *zero-one* matrix are either 0 or 1
  - Representing **False** & **True** respectively.
- The *join* of **A**, **B** (both $m \times n$ zero-one matrices):
  - $\mathbf{A} \vee \mathbf{B} :\equiv [a_{ij} \vee b_{ij}]$
- The *meet* of **A**, **B**:
  - $\mathbf{A} \wedge \mathbf{B} :\equiv [a_{ij} \wedge b_{ij}] = [a_{ij}\, b_{ij}]$

The 1's in **A** join the 1's in **B** to make up the 1's in **C**.

Where the 1's in **A** meet the 1's in **B**, we find 1's in **C**.

Joins and Meets of Zero-One Matrices

Example: Find the join and meet of the zero-one matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: The join of **A** and **B** is

$$\mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

The meet of **A** and **B** is

$$\mathbf{A} \wedge \mathbf{B} = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Boolean Products

- Let **A**=[a_{ij}] be an $m \times k$ zero-one matrix, & let **B**=[b_{ij}] be a $k \times n$ zero-one matrix,
- The *boolean product* of **A** and **B** is like normal matrix $\times$, but using $\vee$ instead of + in the row-column "vector dot product":

$$\mathbf{A} \odot \mathbf{B} = \mathbf{C} = [c_{ij}] = \left[\bigvee_{\ell=1}^{k} a_{i\ell} \wedge b_{\ell j} \right]$$

EXAMPLE 8 Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Solution: The Boolean product $\mathbf{A} \odot \mathbf{B}$ is given by

$$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Boolean Powers

- For a square zero-one matrix **A**, and any $k \geq 0$, the *kth Boolean power of* **A** is simply the Boolean product of k copies of **A**.

$$\mathbf{A}^{[k]} := \underbrace{\mathbf{A} \odot \mathbf{A} \odot \ldots \odot \mathbf{A}}_{k \text{ times}}$$

$$\mathbf{A}^{[0]} := I_n$$

作业

- § 2.6 – 4, 20, 28, 32